Identifying Causal Effects With Latent Treatments: Psychological Experiments as Instrumental Variable Designs

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The R code for the simulation study and the simulated results that reproduce all tables and figures are available at https://github.com/Lukas-Jay/IV-with-Latents.

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Abstract

Psychological experiments rarely manipulate the variable of interest directly, because it is latent. Researchers show a film clip to induce sadness and study how sadness affects an outcome. They confirm that sadness changed with manipulation checks, but then estimate only the effect of the film, not of sadness itself. We target the causal effect of the psychological variable by treating psychological experiments as instrumental variable (IV) designs, with the manipulation as an instrument for a latent treatment. Allowing for person-specific treatment effects and item loadings, we state assumptions that identify this effect. The resulting measurement assumptions are weaker than those psychological studies routinely make, and they identify the estimand psychologists actually care about.

Because a latent treatment has no unit, the estimand is defined only once a scale is fixed. We define two: the effect per latent standard deviation and the effect per unit of a single anchor item. The two differ in usefulness and feasibility. The anchor-scaled estimand depends on the scale of one item; the scale-based estimand is more familiar, but requires more assumptions and does not transfer between experiments. We show how to estimate both effects with off-the-shelf SEM and IV software, and a simulation shows how the estimators behave in realistic samples when single assumptions fail. Because researchers design the manipulation and choose the items, we close with practical implications for both choices.

Keywords: instrumental variables, latent variables, causal inference, experimental manipulation, structural equation modeling

Identifying Causal Effects With Latent Treatments: Psychological Experiments as Instrumental Variable Designs

Psychologists and social scientists are interested in causal effects of changes in latent variables, such as psychological states, traits, or emotions. These latent variables pose two problems for causal inference: latent variables cannot be observed, and, more consequentially, they cannot be manipulated directly in an experiment. For example, to study how sadness X affects a financial decision Y, sadness must be manipulated by proxy, with an emotionally evocative film clip (Lerner et al., 2013). The film clip Z is the experimental manipulation, but it is of no substantive interest in itself: it serves only to shift the target variable X, the emotional state observed via self-reported affect ratings. The same structure recurs whenever a procedure is used to move a latent state: a stereotype-threat framing raises experienced threat (Steele & Aronson, 1995), an induced-compliance procedure provokes dissonance (Festinger & Carlsmith, 1959), the Trier Social Stress Test elevates stress (Kirschbaum et al., 1993), and an effortful task induces ego depletion (Baumeister et al., 1998). In each case, the researcher randomizes Z but not X, so the study is not a classical experiment: observed changes between groups are due to Z, and this randomization does not rule out confounding of the $X \to Y$ relationship by unmeasured variables U. Retrieving the target causal effect (the causal estimand) therefore requires further assumptions about how X, its measurement, and the instrument Z used to shift it are connected.

Knowing these necessary assumptions is vital, because experimenters have some control over design and measurement and can tailor both to make the assumptions plausible. Psychological interventions Z can be characterized as indirect interventions (Junker & Grosz, 2026; Mathur, 2025), which places them in the general framework of instrumental variable (IV) identification, well established in the causal inference literature (Hernán & Robins, 2006; G. Imbens, 2014). IV approaches exploit the randomization of Z even though Z shifts X only partially. Outside of psychology, such a source of exogenous

variation (a lottery, a policy change, or assignment to a treatment arm in a randomized trial) is used to estimate treatment effects even though the treatment itself was not randomized. In this sense, psychological experiments resemble randomized encouragement designs, common in medical trials with non-compliance, where participants can only be encouraged to take a treatment, yet compliance is not enforced.

IV identification has received growing attention in the psychological literature, so far for manifest treatments (Grosz et al., 2024; Maydeu-Olivares et al., 2019; Student, 2026). However, this literature does not account for the treatment itself being measured only indirectly. Latent variables in experiments have so far been studied mainly as outcomes (Fu & Green, 2026a, 2026b; Stoetzer et al., 2025). We build on two features that distinguish the psychological setting from the typical econometric or epidemiological application. First, psychological variables cannot be measured directly: the latent treatment X is observed only through indicators $I_1, \ldots, I_J$, each of which reflects X with error. Second, the instrument Z is chosen by the researcher rather than arising naturally, so it is a nuisance variable introduced only to induce variation in X. Psychologists are not interested in the effect of the film clip, the framing, or the stressor as such; they care about the causal effect of the latent X on the outcome Y.

In this article, we develop a unified set of causal and measurement assumptions under which the average causal effect of a latent, indirectly measured treatment can be identified and estimated from a randomized encouragement design. A latent treatment has no inherent scale, so its effect can be expressed on more than one scale. We define two estimands and provide practical estimation procedures for both. We close with the implications of our results for applied studies.

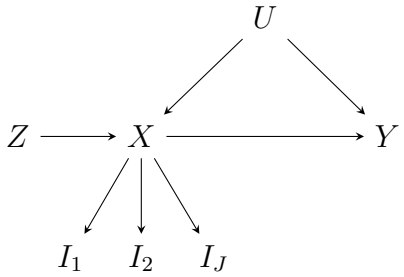
Framework and Notation

We adopt the notation of Structural Causal Models (Pearl, 2009). For each individual i, the data-generating process is defined by structural assignments that specify how each variable is generated from its direct causes and an exogenous error term. Unlike

the linear structural equation models (LSEM) common in psychology, we allow for interactions between the causes of each variable, including the unobserved ones, so causal effects can differ across individuals. We impose functional-form restrictions only where identification requires them.

Figure 1

DAG of an Ideal Psychological Experiment



Note. Z is randomized and affects X. Since X is not randomized, the $X \to Y$ relationship is confounded by U. X is measured reflectively by indicators $I_1, \ldots, I_J$.

Causal Model

The DAG implies the following functions:

$$X_i = f_X(Z_i,\, U_i,\, \epsilon_{Xi}) \tag{1}$$

$$Y_i = f_Y(X_i,\, U_i,\, \epsilon_{Yi}) \tag{2}$$

$$I_{ji} = f_j(X_i,\, U_i,\, \epsilon_{ji}), \qquad j = 1, \ldots, J. \tag{3}$$

U_i contains all unmeasured variables that enter more than one function, and so represents unmeasured confounding such as $X \leftarrow U \rightarrow Y$; ϵ_{Xi}, ϵ_{Yi} and ϵ_{ji} are the remaining exogeneous causes, mutually independent and independent of U. In the running example about the effect of sadness on a financial decision, U consists of chronic financial strain and a dispositional tendency toward low mood, which make participants both sadder and more financially impatient; randomizing the instrument Z does not eliminate U, but it supplies variation in sadness that is independent of it. The functions f_X, f_Y and f_j are the same for everyone, yet the effects are not: U_i and the ϵ terms enter each function, so how X

responds to Z, how Y responds to X, and how each item responds to X can differ from person to person.

Within an SCM, potential outcomes are defined by intervening on the system (Pearl, 2009):

$$X_i(z) = f_X(z, U_i, \epsilon_{Xi}), \qquad Y_i(x) = f_Y(x, U_i, \epsilon_{Yi}), \qquad I_{ji}(x) = f_j(x, U_i, \epsilon_{ji}). \tag{4}$$

$Y_i(x)$ is the value Y_i takes when the structural assignment for X is replaced by the constant x, leaving all other assignments, U_i and the error terms unchanged, and likewise for $X_i(z)$ and $I_{ji}(x)$. These potential outcomes match observed values via consistency, $X_i = X_i(Z_i)$, $Y_i = Y_i(X_i)$ and $I_{ji} = I_{ji}(X_i)$. In psychometric terms, the potential outcome $I_{ji}(x)$, as a function of x, is person i's item response function for item I_j.

Parametric Assumptions

Because Z is binary, the structural equation for X can always be reparameterized without loss of generality as:

$$X_i = \alpha_X + \beta_{Xi} Z_i + \zeta_{Xi} \tag{5}$$

where $\beta_{Xi} = X_i(1) - X_i(0)$ is the individual causal effect of Z on X, $\alpha_X = \mathbb{E}[X_i(0)]$ is the average baseline level of X for $Z = 0$, and $\zeta_{Xi} = X_i(0) - \alpha_X$ is individual i's deviation from it.

Because X is continuous, no such reparameterization exists for $Y_i(x)$ and $I_{ji}(x)$. To identify the target effects, both are assumed to be linear, as stated in Assumptions 4(a) and 6(a):

$$Y_i(x) = \alpha_Y + \beta_{Yi}\, x + \zeta_{Yi} \tag{6}$$

$$I_{ji}(x) = \nu_j + \lambda_{ji}\, x + \zeta_{ji} \tag{7}$$

Here β_{Yi} is person i's treatment effect and λ_{ji} their loading on item j; $\alpha_Y = \mathbb{E}[Y_i(0)]$ and

$\nu_j = \mathbb{E}[I_{ji}(0)]$ are average intercepts, and $\zeta_{Yi} = Y_i(0) - \alpha_Y$ and $\zeta_{ji} = I_{ji}(0) - \nu_j$ are person i's deviations from them. The individual slopes do not depend on the level to which X is set, but they may vary across persons: a unit of sadness may raise one participant's rating by two scale points and another's by one. A coefficient without i is the population mean of the person-specific one, so $\beta_X = \mathbb{E}[\beta_{Xi}]$, $\beta_Y = \mathbb{E}[\beta_{Yi}]$ and $\lambda_j = \mathbb{E}[\lambda_{ji}]$.

The LSEM and the common factor model are the special cases in which $\beta_{Yi} = \beta_Y$ and $\lambda_{ji} = \lambda_j$ for everyone, leaving the ζ terms as the only individual variation. Neither restriction is made here. Confounding by U shows up in Equations (5) and (6) as a correlation between ζ_{Xi} and ζ_{Yi}; it can also make a person's loading λ_{ji} covary with their X_i.

The Target Estimand

The average causal effect (ACE) of moving X by an amount c is

$$\text{ACE}(x, c) = \mathbb{E}[Y_i(x + c) - Y_i(x)], \tag{8}$$

the average difference in Y if X were set to $x + c$ rather than x for everyone in the population. In general, this depends on the baseline level x as well as on the increment c. Under the linear form of Equation (6) it does not depend on the level of x, $\text{ACE}(x, c) = c \cdot \beta_Y$. For example, the change in outcome per unit of induced sadness is just $\beta_Y = \mathbb{E}[\beta_{Yi}]$.

For a manifest treatment, the unit is usually given by the natural unit scale of the variable and we denote the ACE in Equation (8) with $c = 1$ as θ_X, which is equal to β_Y in Equation (6). For a psychological latent variable like sadness, however, the scale of X is not identified, and the causal estimand is not fully specified until the scale has been defined. One possible choice is to define the scale as one standard deviation of the latent treatment, yielding an effect per one latent SD, θ_{SD}. Another is to define the unit through a designated anchor item, yielding an effect per one unit on the anchor scale, θ_{I_0}. An

anchored unit means that the change in financial decision making is expressed per one scale point on a designated sadness rating. These two choices require different identification assumptions, because they define different causal estimands. This scale problem is central in the present setting, because the instrument Z affects the variance of X. We return to this distinction in the section on the anchor-metric estimand.

Note that the causal effect of the experimental manipulation without reference to X, $\text{ITT(z)} = \mathbb{E}[Y_i(z{=}1) - Y_i(z{=}0)]$, is called the intention-to-treat (ITT) effect. By virtue of the randomization of Z, it equals the difference in group means $\mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0]$ and is directly estimable (e.g., with a t-test). However, the ITT is not our target estimand. It conflates the target θ_X with the possibly heterogeneous effect of Z on X (Mathur, 2025).

The Manifest-Treatment Case

We first illustrate the IV-design setup if X were manifest and observed directly, that is, Figure 1 without the indicators I_j. In the running example, this amounts to treating experienced sadness X as if it were observed directly and using Z as an instrument that allows to identify the ACE on the financial decision Y. Identification with manifest treatment rests on four assumptions. The first three are foundational: if they hold, Z is a valid instrument, which allows for the identification of bounds for the causal estimand. To achieve point-identification of the effect, a fourth assumption regarding effect moderation is necessary (Hartwig et al., 2023; Wang & Tchetgen Tchetgen, 2018). We extend these assumptions later to the measurement model of a latent X.

The Three Instrumental Conditions

Assumptions 1–3 for a valid instrument Z can be inferred from a DAG like Figure 1.

Assumption 1 (Relevance)**.** *The instrument Z has a non-zero average causal effect on the treatment X:*

$$\beta_X = \mathbb{E}\Big[X_i(1) - X_i(0)\Big] = \mathbb{E}[\beta_{Xi}] \neq 0$$

Assumption 2 (Exclusion Restriction)**.** *All directed paths from Z to Y in the causal graph*

pass through X, so that Z and Y are independent given the treatment and the unmeasured common causes:

$$Z_i \perp\!\!\!\perp Y_i \mid X_i, U_i$$

Equivalently, neither Z nor any descendant of Z other than X enters the structural function f_Y as an argument. Informally, Z affects Y only by shifting X.

Assumption 3 (Randomization)**.** *The instrument Z is jointly independent of all unmeasured variables and error terms:*

$$Z_i \perp\!\!\!\perp \left(U_i, \epsilon_{Xi}, \epsilon_{Yi}, \epsilon_{1i}, \ldots, \epsilon_{Ji}\right)$$

Given the linear forms in Equations (6) and (7), every person-specific quantity introduced so far is a function of U_i and the ϵ terms alone. Assumption 3 therefore makes all of them independent of Z.

Collectively, these three assumptions imply that Z is an instrumental variable, which allows the derivation of bounds on the estimand (Hernán & Robins, 2006).[1] In a randomized experiment, Assumptions 1 and 3 hold by construction: the instrument is randomized and is chosen precisely because it is expected to move the construct. Assumption 2 is the most problematic in our context. Graphically, it is violated if a direct path $Z \to Y$ exists, or any path through an unobserved intermediate M such that $Z \to M \to Y$. In the running example, the sadness film must not affect financial choice except by making participants sad. If the clip also induces physiological arousal, boredom, or time pressure, and any of these shifts financial decisions, the exclusion restriction fails, even though the manipulation "worked" by raising reported sadness.

[1] The assumptions stated here are slightly stronger than necessary, tailored to psychological experiments for clarity. For example, Z need not be randomized or causally efficacious itself (Hernán & Robins, 2020; Pearl, 2009). Nor does a manifest X require Z to be independent of the item errors ϵ_{ji}; we include it here because randomization guarantees it anyway and the latent case introduced later relies on it.

The Fourth Point-Identifying Condition: Homogeneity

By itself, a valid instrument Z does not suffice to point-identify the causal estimand. What the instrument recovers is at best a weighted average of individual causal effects rather than the ACE in the whole population. Point identification of the ACE requires an additional assumption about effect homogeneity. The more commonly discussed alternative in econometrics is to forgo the ACE and instead target the local average treatment effect (LATE), the causal effect among the subgroup of *compliers*, identified under a monotonicity assumption (Angrist et al., 1996). The LATE is typically not the estimand psychologists are interested in (see Box 1), so we take the homogeneity route instead. Psychologists know this route far less well than the LATE framework, which is why we develop it in detail here. It also shows an instructive contrast: the econometric IV literature and psychometric modelers are willing to make quite different modeling assumptions.

For intuition, recall that IV identification works by inducing variation in X through an intervention on Z. Because Z is randomized, this induced variation is independent of any $X \to Y$ confounding. The exclusion restriction then licenses the claim that all observed differences in Y between levels of Z operate through differences in X. Conceptually, however, we only learn directly about those individuals whose X actually changes in response to Z—if X were binary, these are the *compliers* with respect to the instrument. Participants not moved by the film clip contribute no variation in sadness, so the IV signal reflects only those whose sadness actually changes.

Box 1: Why Not the LATE?

In the econometrics literature, the well-known alternative to Assumption 4 is monotonicity (G. W. Imbens & Angrist, 1994). Assumptions 1–3 together with monotonicity identify the local average treatment effect (LATE), the causal effect in the subgroup of compliers. Monotonicity asserts that the instrument shifts treatment in one direction only: for every individual, being in the experimental group can raise X or leave it unchanged, but never lower it.

We do not pursue the LATE, because it is not a useful estimand in the settings we consider. There are at least three reasons.

First, the LATE depends on the instrument: different instruments induce different rates of compliance and thus different estimands. A sadness film and a sad autobiographical writing task move different people, so each identifies the effect of sadness on decision making in a different subpopulation. Psychologists rarely care about effects tied to one particular manipulation; they use instruments to shift a target construct and want the effect of that construct on the outcome. For the same reason, mediation effects of Z through X are not of substantive interest (Mathur, 2025).

Second, the complier population is itself not identified: without further assumptions, no individual can be classified as a complier. For continuous treatments such as psychological constructs, "compliers" are not even a well-defined group. With continuous X, the LATE becomes a weighted average of heterogeneous effects whose weights depend on the instrument in ways that are hard to interpret substantively (Angrist et al., 1996).

Third, monotonicity is not plausible in psychological settings. In medical trials it is defensible, because patients can typically access the treatment only if assigned to the treatment group. In psychological experiments, participants in both groups can move their latent state up or down, and it is unreasonable to assume that everyone responds to the instrument in the same direction or not at all. A film clip meant to induce sadness may leave some participants unmoved and strike others as absurd, cheering them up.

Strict Homogeneity

There are two possible conditions, each of which allows us to generalize immediately from the compliers to the population. Under the first, everyone responds to Z in exactly the same way, that is, the effect of Z on X is constant and additive across individuals: the film clip makes every participant sadder by the same amount. The instrument-induced variation in X is then homogeneous and there is no relevant non-complier subgroup to speak of. Under the second, participants differ in how sad the clip makes them, but a given increase in sadness changes the financial decision Y by the same amount for everyone. The complier-specific effect then coincides with the population ACE, and the IV estimate can be interpreted as the ACE.

Either condition suffices, and both are arguably strong and unrealistic, which is why the IV literature opted for different ways (see Box 1). Yet both conditions hold simultaneously in an LSEM without interactions. An LSEM takes Y to be linear in X and additionally restricts β_{Xi} and β_{Yi} to be constant across individuals. This is a peculiar assumption in light of Figure 1, which posits that unmeasured confounders U exist and jointly influence both X and Y: financial strain may make participants sadder and more impatient on average, but it is not allowed to make them more responsive to the film or more swayed by their own sadness. Many psychologists readily make these assumptions due to modeling convencience, because the standard LSEM encodes them by default.

The NOSH Condition

While LSEMs are popular in psychological science, they are neither intuitive nor plausible through the lens of counterfactual causal inference. We typically expect the effect of an intervention Z on X to vary substantially across individuals, and likewise expect the causal effect of X on Y to be modified by person-level characteristics. In that sense, the LSEM encodes a very special case of homogeneity: all effect modification by unmeasured variables is ruled out by construction.

Recent work has shown that this "constant effect" assumption is stricter than

necessary for point identification of the ACE (Hartwig et al., 2023; Wang & Tchetgen Tchetgen, 2018). A more general condition is the following:

Assumption 4 (No Simultaneous Heterogeneity (NOSH))**.** *This assumption has two parts (Hartwig et al., 2023):*

(a) Linearity of the treatment effect. *The individual potential outcomes $Y_i(x)$ are linear in x, as in Equation* (6)*:*

$$Y_i(x) = \alpha_Y + \beta_{Yi}\, x + \zeta_{Yi}$$

where β_{Yi} and ζ_{Yi} may vary freely across individuals but do not depend on the level x at which the change occurs.

(b) Uncorrelated effect heterogeneity. *The individual-level instrument effect β_{Xi} and the individual-level treatment effect β_{Yi} are uncorrelated:*

$$\mathrm{Cov}(\beta_{Xi},\, \beta_{Yi}) = 0$$

Part (a) means that the individual β_{Yi} may vary between persons, but must be constant for all levels of x within persons: for a given person, going from mildly to moderately sad shifts Y by the same amount as going from moderately to very sad. Part (b) means that the unmeasured variables making some individuals more responsive to the instrument are not systematically associated with the unmeasured variables making the treatment more or less effective for them. [2] For example, it may fail if sadness is induced with a film depicting financial hardship: financially strained participants are plausibly more moved by such a film (a larger β_{Xi}) and more prone to let the resulting sadness sway their monetary choices (a larger β_{Yi}), so $\mathrm{Cov}(\beta_{Xi}, \beta_{Yi}) > 0$. NOSH thus generalizes

[2] Part (b) is stated in covariance form because that is the condition the binary randomized-Z Wald argument actually uses (see Appendix A). Stronger versions, like mean independence $\mathbb{E}[\beta_{Yi} \mid \beta_{Xi}] = \mathbb{E}[\beta_{Yi}]$ or full independence $\beta_{Yi} \perp\!\!\!\perp \beta_{Xi}$ are typically invoked when the instrument is continuous or not randomized. They imply part (b) but are stronger than what is needed here.

constant-effect assumptions: it allows the effects to vary across individuals but restricts how the individual effects β_{Xi} and β_{Yi} are related.

The Wald Ratio

If Assumptions 1–4 (IV conditions plus NOSH) hold and Z, X, and Y are all manifest, the Wald ratio (Wald, 1940) identifies the desired causal estimand (proof in Appendix A). For a binary instrument Z, the Wald ratio is

$$\beta_{\text{IV}}(X) = \frac{\text{Cov}(Z, Y)}{\text{Cov}(Z, X)} = \frac{\mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0]}{\mathbb{E}[X \mid Z = 1] - \mathbb{E}[X \mid Z = 0]}, \tag{9}$$

the reduced-form effect of Z on Y divided by the effect of Z on X. Under Assumptions 1–4, $\beta_{\text{IV}}(X) = \beta_Y = \theta_X$. Implementations to estimate this ratio, via two-stage least squares with the `ivreg` package (Fox et al., 2026) and as a path model in `lavaan`, are given in Appendix B.

The Latent-Treatment Case

Psychological variables like sadness are never observed directly but measured via reflective indicators $I_1, \ldots, I_J$, such as self-reported emotion ratings. If X is latent, the denominator of Equation (9), $\text{Cov}(Z, X)$, is unobservable. The numerator, $\text{Cov}(Z, Y)$, remains observable, but the first stage must now be expressed through the manifest indicators of X. Because X is observed only through these indicators, its unit is not intrinsic but fixed by convention. This is a central problem in the present setting. Causal effects are defined by interventions on X, so their definition depends on the scale of X: each choice of unit yields a different causal estimand.

A common choice is to standardize X, either explicitly by imposing $\text{Var}(X) = 1$ or implicitly by reporting effects per 1 SD of X (see Stoetzer et al., 2025). In our case, however, the SD of X depends on the strength of the instrument Z, a problem we return to below. Let $\sigma_X = \sqrt{\text{Var}(X)}$ denote the standard deviation of X, pooled over both

conditions Z. The standardized latent estimand is

$$\theta_{\text{SD}} = \mathbb{E}[Y_i(x + \sigma_X) - Y_i(x)] = \sigma_X \, \beta_Y. \tag{10}$$

Another possibility is to fix the scale through the loading of a single item, an *anchor*, so that the effect is reported per one unit on that item's metric. The anchor is one of the indicators $I_1, \ldots, I_J$, relabeled to I_0. One unit of I_0 stands for a fixed amount of X only if the anchor responds linearly to the trait, which is Assumption 6(a) below. The anchored estimand is

$$\theta_{I_0} = \mathbb{E}\left[Y_i\left(x + \frac{1}{\lambda_0}\right) - Y_i(x)\right] = \frac{\beta_Y}{\lambda_0}, \tag{11}$$

where $\lambda_0 = \mathbb{E}[\lambda_{0i}] \neq 0$ is the anchor's average loading: an item with a zero average loading carries no information about X and cannot lend it a unit.

The remainder of this section explores identification assumptions for these quantities to recover the causal effect of the latent treatment. We start with the latent scale estimand and then show that the anchor estimand is identifiable under fewer assumptions.

Identification Assumptions for the Standardized Latent Estimand

Because X is latent, neither $\text{Cov}(Z, X)$ nor σ_X can be observed, and both have to be recovered from the indicators. For any indicator I_j with average loading $\lambda_j = \mathbb{E}[\lambda_{ji}] \neq 0$,

$$\theta_{\text{SD}} = \lambda_j \sigma_X \, \beta_{\text{IV}}(I_j) \tag{12}$$

if the following assumptions hold (proof in Appendix A). The Wald ratio $\beta_{\text{IV}}(I_j)$ is Equation (9) with the observed indicator I_j in place of the latent X, so it can be estimated from observed data. The standardized loading $\lambda_j \sigma_X$, the loading of I_j when $\text{Var}(X)$ is fixed to one, is not, and has to be recovered from the covariances among the indicators.

Assumptions for the Instrument

Because IV compares the groups induced by the instrument Z, X must be measured in a similar manner in both of them. This implies that there should be no effect of Z on any indicator I_j except through X, which is essentially another exclusion restriction (Stoetzer et al., 2025).

Assumption 5 (Measurement Exclusion)**.** *All directed paths from Z to any indicator I_j in the causal graph pass through X, so that*

$$Z_i \perp\!\!\!\perp I_{ji} \mid X_i, U_i \qquad \text{for every } j.$$

Equivalently, neither Z nor any descendant of Z other than X enters the structural functions f_j of the items I_j as an argument.

Under Assumption 5, the structural functions f_j in Equation (3) contain no Z, and the instrument reaches the indicators only by shifting X. Each potential outcome $I_{ji}(x) = f_j(x, U_i, \epsilon_{ji})$ in Equation (4) is therefore a function of U_i and ϵ_{ji} alone, and these are independent of Z by randomization (Assumption 3). Hence $I_{ji}(x) \perp\!\!\!\perp Z$ for every x.

As in the manifest case, an exclusion restriction alone is not enough for point identification. We require some form of homogeneity to ensure that differences in the items between the conditions of Z are proportional to the average change in X. This can fail in two ways. First, if a person's loading depends on the level of X, the loadings differ between conditions, because Z shifts X. Second, if the people whom Z moves most also have the steepest loadings, they dominate these differences. The next assumption rules out both by transferring the two parts of Assumption 4 from the outcome equation to the item equations.

Assumption 6 (Item NOSH)**.** *This assumption has two parts, mirroring Assumption 4:*

(a) Linearity of the item response. *Person i's item response function $I_{ji}(x)$ is linear in x,*

as in Equation (7)*,*

$$I_{ji}(x) = \nu_j + \lambda_{ji}\, x + \zeta_{ji} \quad \textit{for all } j,$$

where λ_{ji} *and* ζ_{ji} *may vary freely across individuals but do not depend on the level* x *at which the item responds.*

(b) Uncorrelated loading heterogeneity. *The loading is uncorrelated with first-stage compliance,*

$$\mathrm{Cov}(\lambda_{ji},\, \beta_{Xi}) = 0 \quad \textit{for all } j.$$

Assumption 6(a) requires linear item response functions: the person-specific loadings do not depend on the level of X. Assumption 6(b) is the measurement-side analog of Assumption 4(b): just as $\mathrm{Cov}(\beta_{Xi}, \beta_{Yi}) = 0$ prevents the outcome slope from collapsing to a compliance-weighted average, $\mathrm{Cov}(\lambda_{ji}, \beta_{Xi}) = 0$ prevents the loading from doing so. Both parts concern measurement, but they are needed for the instrumental-variable step itself rather than for measuring X correctly (Appendix A).

Under Assumption 6(a), the observed indicators take the constant-loading form of factor analysis. With the average loading λ_j, individual loading heterogeneity $\eta_{ji} = \lambda_{ji} - \lambda_j$, and composite item residual $\delta_{ji} = \eta_{ji} X_i + \zeta_{ji}$,

$$I_{ji} = \nu_j + \lambda_j X_i + \delta_{ji}. \tag{13}$$

The loading heterogeneity now sits in the residual. Observed covariances cannot separate η_{ji} from ζ_{ji} within δ_{ji}, yet η_{ji} means that persons differ in the slopes of their item response functions $I_{ji}(x)$. Assumption 6(b) ensures that these differences in slope do not matter for the instrumental-variable step.

The assumptions so far identify the effect per unit of each item, $\beta_{\mathrm{IV}}(I_j) = \beta_Y / \lambda_j$. To obtain θ_{SD}, this ratio has to be multiplied by the standardized loading $\lambda_j \sigma_X$, which is recovered from the covariances among the indicators. This works only if δ_{ji}, which now

contains the term $\eta_{ji}X_i$, still behaves like an ordinary factor-analytic residual, uncorrelated across items and with X. To ensure this, we need the measurement assumptions explicated below.

Assumptions for the Factor Model

The two assumptions below do not involve Z. They are the standard conditions a factor model needs to recover the standardized loading $\lambda_j\sigma_X$ from the covariances among the indicators, and with it the estimand θ_{SD}.

The main advantage of instrumental variables is that they tolerate confounding of the form $X \leftarrow U \rightarrow Y$ even when U is unobserved. With latent treatments, this protection extends to confounding between an indicator and the outcome, $I_j \leftarrow U_j \rightarrow Y$: because Z is independent of U_j, the Wald ratio $\beta_{\mathrm{IV}}(I_j)$ is unaffected. Identifying the scale of X, however, requires the item-level confounders U_j to be mutually independent. Otherwise they induce correlations between indicators that X does not generate, and the measurement part is misspecified. This condition is easiest to check graphically (Figure 2).

Assumption 7 (Local Independence)**.** *There is no residual covariance across indicators beyond the covariance generated by the latent treatment*

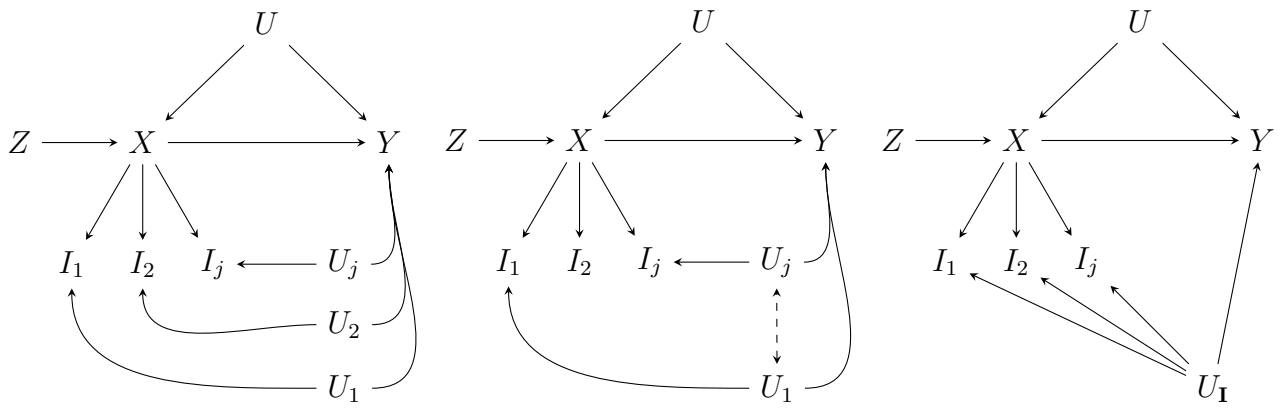
$$\mathrm{Cov}(\delta_{ji},\, \delta_{ki}) = 0 \quad \textit{for all } j \neq k$$

This ensures that the shared variance among indicators is attributable to the latent treatment X. Individual items may be influenced by item-specific causes that also affect Y (i.e., $I_j \leftarrow U_j \rightarrow Y$ paths are permitted), provided these causes are mutually independent across items.

The final assumption is a measurement-validity requirement: there should be no $X \leftarrow U_j \rightarrow I_j$, because all changes we measure for each I_j should be caused by changes in X.

Figure 2

When Do Confounders Violate Assumption 7?



(a) *Assumption 7 holds* (b) *Assumption 7 violated* (c) *Assumption 7 violated*

Assumption 8 (No X–I Confounding)**.** *The composite item residual is uncorrelated with the latent trait,*

$$\mathrm{Cov}(X_i,\, \delta_{ji}) = 0 \quad \textit{for all } j,$$

which rules out backdoor confounding $X \leftarrow U_j \rightarrow I_j$.

Assumption 8 can be violated if a cause of X also affects an item's intercept or its loading. In the notation of Equation (13), the two are $\mathrm{Cov}(X_i, \zeta_{ji}) \neq 0$ and $\mathrm{Cov}(X_i, \eta_{ji}X_i) \neq 0$. Assumption 8 bundles them into a single restriction on δ_{ji}, but they are logically independent and either one alone violates it. Assumptions 5, 7, and 8 can each fail even when Z is a valid instrument for X (Figure 3).

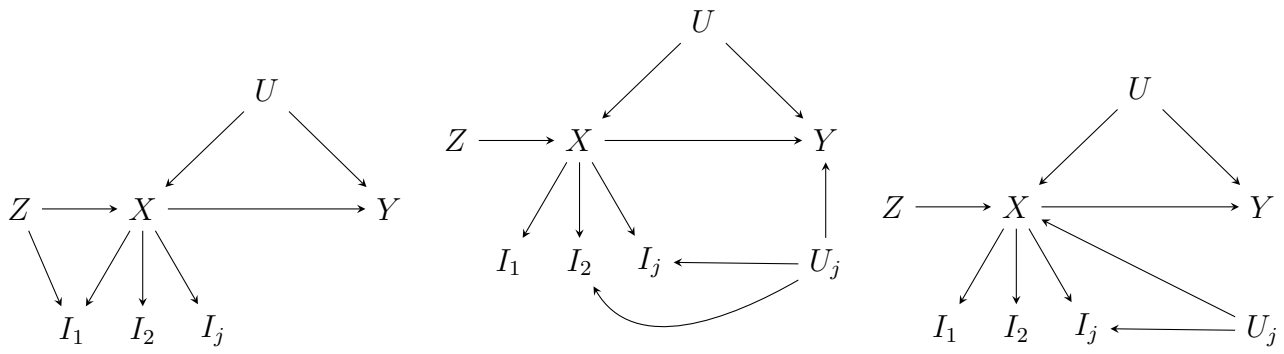
Assumptions 7 and 8 secure a valid scale: they ensure that the covariance structure of the indicators reflects the latent trait rather than item-specific contamination, so that the loadings, and through them the latent variance, can be recovered. Then, under Assumptions 1–8, Equation (12) holds: $\lambda_j \sigma_X \, \beta_{\mathrm{IV}}(I_j) = \sigma_X \beta_Y = \theta_{\mathrm{SD}}$ (proof in Appendix A).

The Anchor-Metric Estimand: Fewer Assumptions, Transportable Across Z

The standardized latent estimand is natural from a psychometric perspective, but it is not always the most useful causal estimand. Its unit is fixed by the variance of X, so the

Figure 3

Z Is a Valid Instrument, but θ_{SD} Is Not Identified



(a) *Assumption 5 violated* (b) *Assumption 7 violated* (c) *Assumption 8 violated*

estimand depends on the distribution of the latent treatment. This becomes a problem when effects are transported to settings in which X has a different variance: to populations that vary more in their sadness, or to designs in which another film clip or a different Z shifts sadness more strongly. A sadder film makes one SD of sadness a larger dose, and so yields a larger reported effect. Comparability across studies is a central aim of experimental design, and study-specific scales are a recognized obstacle to it (Fu & Green, 2026a).

Transporting effects requires a scale for X that is invariant across instrument conditions. An anchor item provides one: Because Z does not enter the item's structural function (Assumption 5) and the item responds linearly to X (Assumption 6(a)),[3] "one unit of I_0" corresponds to the same change in X regardless of which Z is used. The anchor-metric estimand is then identified by a Wald ratio in which the anchor takes the place of X. Substituting the item response function of Equation (7) and the first-stage representation of Equation (5) (Appendix A),

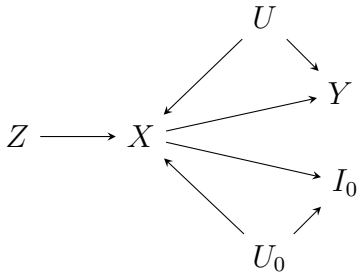
$$\beta_{\text{IV}}(I_0) = \frac{\text{Cov}(Z, Y)}{\text{Cov}(Z, I_0)} = \frac{\mathbb{E}[\beta_{Yi}\beta_{Xi}]}{\mathbb{E}[\lambda_{0i}\beta_{Xi}]} \tag{14}$$

[3] If the anchor were nonlinear, the per-unit effect would vary across the range of X. The Wald ratio would then return an average weighted by where on φ the instrument happens to move respondents. Like the LATE (Box 1), this average depends on the instrument, which defeats the purpose of the anchor metric.

Under Assumptions 1–6, the ratio in Equation (14) equals $\beta_Y/\lambda_0 = \theta_{I_0}$ (proof in Appendix A). The anchor strategy does not need the two assumptions that secure the scale of a factor model.[4] The reason is that the anchor is essentially just another outcome of X, and the only indicator that has to be considered (Figure 4). Because it is the only indicator, local independence (Assumption 7) is not needed: the ratio uses no inter-item covariance. Because it is just another outcome, the ratio also tolerates X–I confounding (Assumption 8). Confounding $X \leftarrow U_0 \rightarrow I_0$ plays the same role as $X \leftarrow U \rightarrow Y$ in ordinary IV: whether it acts on the anchor's intercept or its loading, it leaves $\mathrm{Cov}(Z, I_0)$ untouched, because randomization keeps U_0 independent of Z and Assumption 6(b) keeps the anchor's loading uncorrelated with compliance.

Figure 4

The Anchor Item as Another Outcome of X



Note. This DAG plus NOSH conditions 4 and 6 allow identification of the anchor estimand.

Beyond measurement, transporting effects also requires that the IV estimand is invariant across instrument conditions. Randomization and the two uncorrelated-heterogeneity conditions (Assumptions 3, 4(b), and 6(b)) ensure that each study's Wald ratio recovers the anchor-metric effect θ_{I_0} rather than an average weighted by how strongly Z moves each person. Average compliance β_X appears in both numerator and denominator and cancels, so the strength of the instrument, which inflates θ_{SD}, never affects θ_{I_0}. The necessary conditions for transporting effects are already in place, so any item with a nonzero average loading can fix the scale of X and make it portable across

[4] In practice, additional measurement assumptions can still be imposed so that the other indicators enter a latent IV model and improve efficiency, as is done in the simulation study.

instruments (see also Fu & Green, 2026a, 2026b). With a "how sad do you feel" item as I_0, the estimand becomes the change in the financial decision associated with a one-point increase on that item. Because the rating registers sadness in the same way regardless of Z, this per-item effect is comparable across studies that induce sadness with a different film clip, or with any other instrument Z, for which Assumptions 1–6 hold. Table 1 summarizes the identifying assumptions of the previous sections.

Table 1
Identifying assumptions for the manifest, standardized-latent, and anchor-metric estimands

	Manifest θ_X	Standardized θ_{SD}	Anchor θ_{I_0}
Resulting estimand			
Unit of the effect	1 unit of X	1 SD of X	1 unit of I_0
Identified quantity	β_Y	$\sigma_X\,\beta_Y$	β_Y/λ_0
Portable across instruments Z	✓	— [c]	✓[c]
Assumptions			
1 Relevance	✓	✓	✓
2 Exclusion restriction	✓	✓	✓
3 Randomization	✓	✓	✓
4(a) Linearity of Y in X	✓	✓	✓
4(b) $\text{Cov}(\beta_{Xi}, \beta_{Yi}) = 0$	✓	✓	✓
5 Measurement exclusion	—	✓[a]	✓[b]
6(a) Linearity of the item response	—	✓[a]	✓[b]
6(b) Uncorrelated loading heterogeneity, $\text{Cov}(\lambda_{ji}, \beta_{Xi}) = 0$	—	✓[a]	✓[b]
7 Local independence	—	✓[a]	—
8 No X–I confounding, $\text{Cov}(X_i, \delta_{ji}) = 0$	—	✓[a]	—

Note. Resulting estimand: ✓ = yes; — = no. Assumptions: ✓ = required; — = not required. [a] At all J indicators entering the factor model. [b] At the anchor item I_0 only. [c] The SD metric is fixed by $\text{Var}(X)$ in the sample at hand, which depends on how far the particular instrument moves the trait; the anchor metric is fixed by the item and does not.

Simulation

A simulation study supplements the theoretical results and serves three purposes. First, it confirms that both estimands are estimated without bias at a large sample size when theory predicts identification (see Table 1) and an appropriate estimator is chosen. Second, it compares the estimators at a more realistic sample size, in scenarios where both estimands are identified and in scenarios where they are not. Third, it shows how the estimators perform for both estimands when a single item, or the anchor item itself, is affected by assumption violations.

Setup and Scenarios

We evaluate how accurately three estimators recover the causal estimands in fourteen scenarios (Table 2, Figure C1), each with 10,000 replications at a small ($n = 250$) and a large ($n = 2{,}000$) sample size. In the baseline scenario, observations are drawn from an SCM corresponding to Figure 1, with ten indicators and the linear functional forms of Equations (5)–(7). X and Y are confounded ($X \leftarrow U \rightarrow Y$), but no structural or functional assumption is violated. The instrument Z is strong: its effect on X is $d \approx 0.66$, comparable to a successful manipulation check. The average causal effect of X on Y is large (standardized $\beta \approx .56$). Nine of the ten items are ordinary items with an average standardized loading of .70; the tenth is the anchor, with a standardized loading of .75 (Table C1). All simulations were run in R (R Core Team, 2026).

Each remaining scenario uses the same structure and adds confounding to test one or more assumptions. Table 2 describes the eight main scenarios. The violations are moderate and tuned to produce biases of similar size (Table C1; Figure C1 draws the causal structure of each scenario). The last two columns of Table 2 show whether theory predicts each estimand to be identified. In the first three scenarios, the prediction is the same for both. In the last five, the violation affects only the ordinary items, so θ_{I_0} remains identified but θ_{SD} does not.

In all eight main scenarios, the anchor item is not affected by the confounding a

Table 2
Simulation scenarios

No.	Scenario	Violates	Realization in the DGP	θ_{I_0}	θ_{SD}
1	All clean	none	Ideal DGP as in Figure 1.	yes	yes
2	Indep. $U_j \to I_j, Y$	none	Each item has its own $U_j \to I_j, Y$.	yes	yes
3	NOSH violated	4(b)	Moderators of $Z \to X$ and $X \to Y$ are correlated.	no	no
4	Indep. $U_j \to I_j, X$	8	Each item has its own $U_j \to X, I_j$.	yes	no
5	DIF by Z	5 (7, 8)	All ordinary items show weak DIF by Z in intercept and loading; the anchor is unaffected.	yes	no
6	Shared $U \to I, Y$	7	All ordinary items share one confounder $U \to I_j, Y$.	yes	no
7	Two more factors	7	Items load on two additional orthogonal factors, three items on the first and seven on the second.	yes	no
8	Item-NOSH violated	6(b) (8)	The loadings of all ordinary items are moderated by the variable that drives compliance, so $\text{Cov}(\lambda_{ji}, \beta_{Xi}) \neq 0$; the anchor is unaffected.	yes	no

Note. DGP = data-generating process; Indep. = independent; NOSH = no simultaneous heterogeneity; DIF = differential item functioning. Names match the figures; "yes" means theory predicts the estimand to be identified. Assumptions in parentheses are broken as a by-product of the first: a compliance-moderated loading also correlates the item's residual with X, and DIF by Z makes Z a common cause of X and every item.

scenario adds. Scenarios 2 and 4–8 also come in a second version with a contaminated anchor: a tenth ordinary item replaces the clean anchor, so all indicators are contaminated. These versions check the result that θ_{I_0} remains identified when the anchor violates Assumption 7 or 8, but not when it violates Assumption 5 or 6.

Estimands and Estimators

Three estimators (SEM-IV, Two-step FA-IV, and Anchor-Wald) target two estimands, θ_{I_0} and θ_{SD} (Equations (11) and (10)). Latent models are fitted with `lavaan`

(Rosseel, 2012) and regressions with `ivreg` (Fox et al., 2026); Appendix B describes the estimators in detail.

SEM-IV estimates the measurement model and the structural IV equations simultaneously in a single SEM in `lavaan`. The model specifies no correlated item errors and is therefore susceptible to measurement violations. It does specify the IV structure correctly, with an `X ~~ Y` residual correlation that absorbs the confounding by $X \leftarrow U \rightarrow Y$.

Two-step FA-IV first fits a confirmatory factor analysis (CFA) in `lavaan` and extracts Bartlett factor scores for X, which are conditionally unbiased given the factor. It then treats these scores as a manifest treatment in a standard IV analysis, a two-stage least squares regression with `ivreg`. Because Y is not part of the CFA, the `I ~~ Y` residual covariances stay unconstrained, whereas SEM-IV fixes them to zero.

Anchor-Wald estimates the Wald ratio of Equation (14) by plugging in the sample means of Y and I_0 in the groups defined by Z. The other two estimators use all indicators and explicitly model the relationship between X and its items. Anchor-Wald uses only one indicator, the anchor, and on its own estimates only θ_{I_0}.

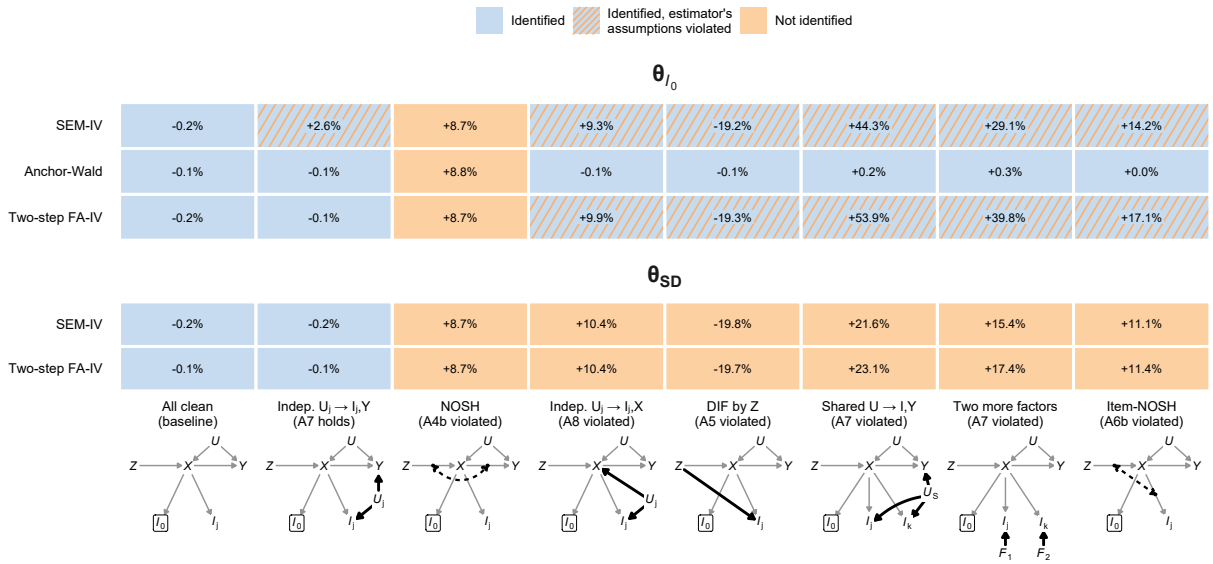
Results

Figure 5 shows whether the three estimators are biased with a clean anchor at a large sample size of $n = 2{,}000$. Anchor-Wald is unbiased for θ_{I_0} in every scenario in which theory predicts identification, and biased in the NOSH scenario, in which it does not. SEM-IV and two-step FA-IV are biased where unmodeled confounding in the item relations violates their model assumptions. SEM-IV is also biased for θ_{I_0} under independent $U_j \rightarrow I_j, Y$ confounding, although the estimand is identified in theory, because the model is misspecified: the SEM fixes the residual covariances between the items and Y to zero, and this confounding violates that restriction. The misfit shrinks the unit of X relative to the anchor, and θ_{SD}, which does not depend on that unit, stays unbiased. Two-step FA-IV is agnostic to this confounding and unaffected. Tables C3 and C5 report median bias, RMSE,

and coverage for these conditions.

Figure 5

Median Bias With a Clean Anchor



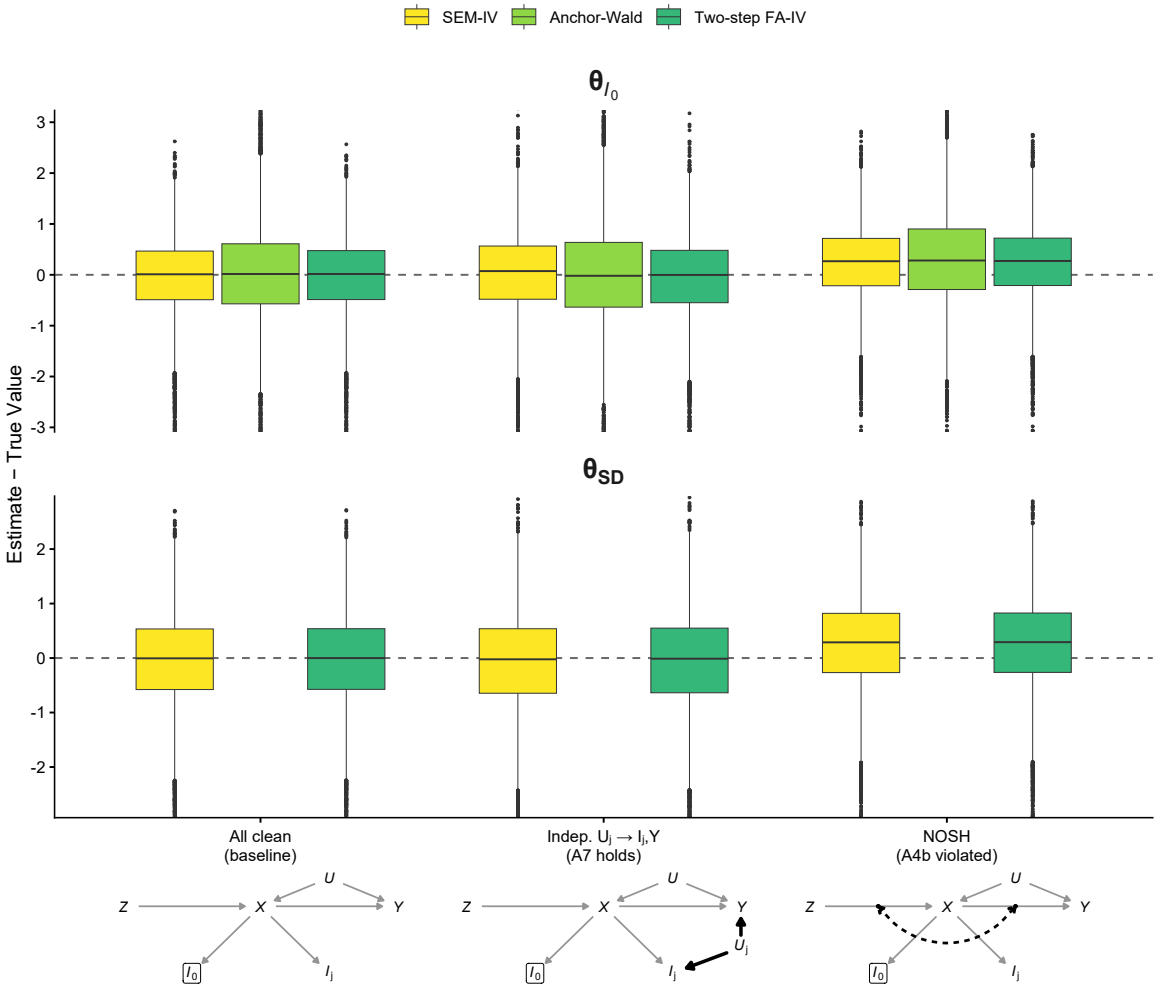
Note. $n = 2{,}000$. Cells show the relative median bias, the median of (estimate − true value) / true value. Shading marks whether theory predicts the estimand to be identified. Hatching marks cells in which the estimand is identified but the estimator relies on assumptions the scenario violates. The graph under each column marks the violating path (black) against the assumed model (gray); the boxed node is the anchor. The anchor item I_0 is not affected by any violation.

Figures 6 and 7 show the same scenarios in more detail at $n = 250$, a sample size more realistic for experimental psychological research (Tables C2 and C4 report median bias, RMSE, and coverage). Figure 6 shows the first three scenarios, in which both estimands are identified or neither is. Anchor-Wald, which rests on the anchor item alone, has a larger variance than the other two estimators. Under the NOSH violation, all three estimators are biased.

Figure 7 shows the last five scenarios, in which θ_{I_0} is identified but θ_{SD} is not. As expected, Anchor-Wald stays unbiased throughout. SEM-IV and two-step FA-IV are biased on the I_0 metric although the estimand is identified, because both learn from the contaminated items.

Figure 6

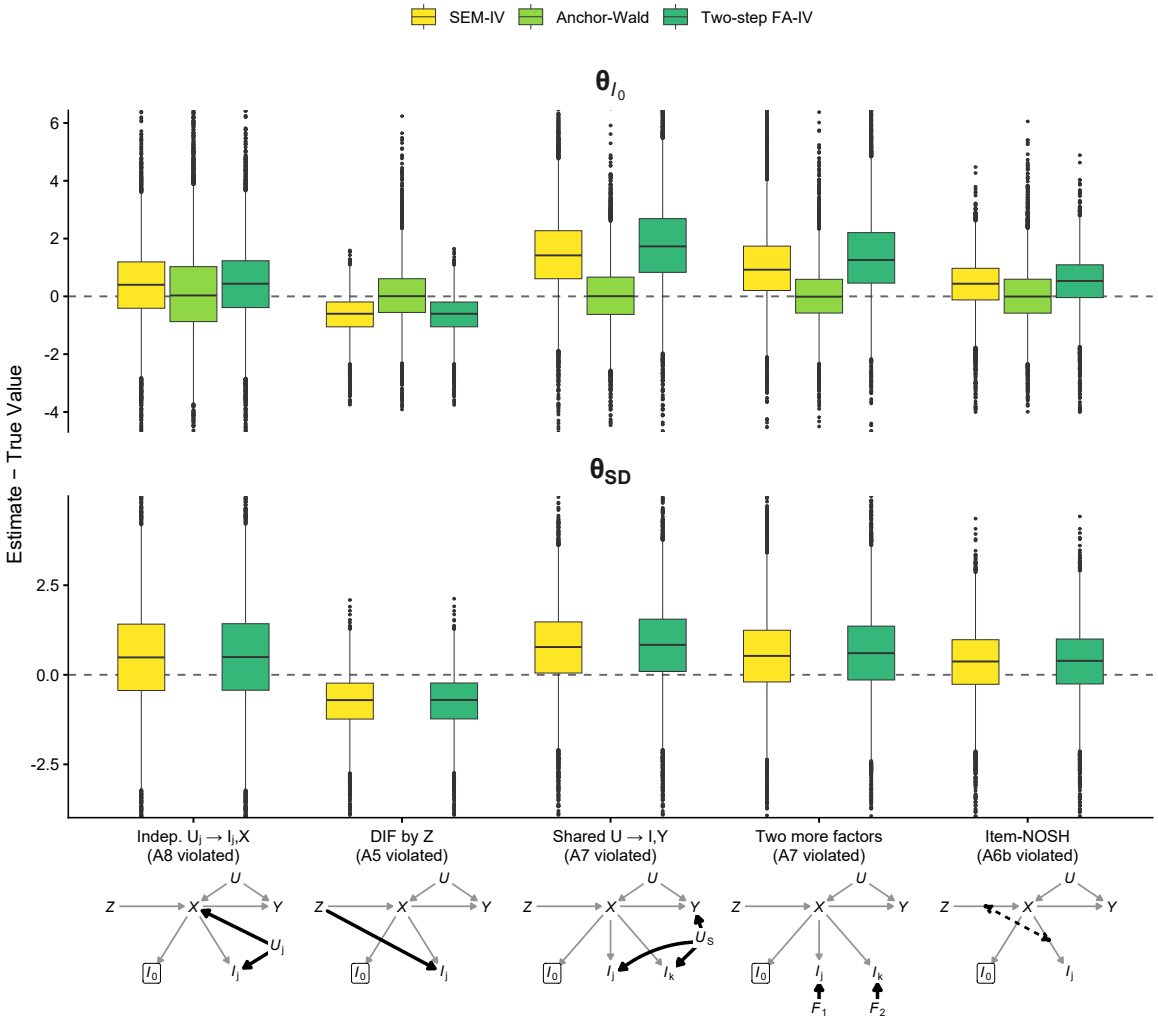
Bias With a Clean Anchor: Both Estimands Are Identified or Neither Is



Note. Bias on the I_0 metric (top) and on the SD scale (bottom), $n = 250$. Both estimands are identified in the clean baseline and under independent $U_j \to I_j, Y$ confounding, and neither is under a NOSH violation (Assumption 4(b)). Anchor-Wald has no SD-scale counterpart.

Figure 7

Bias With a Clean Anchor: Only θ_{I_0} Is Identified



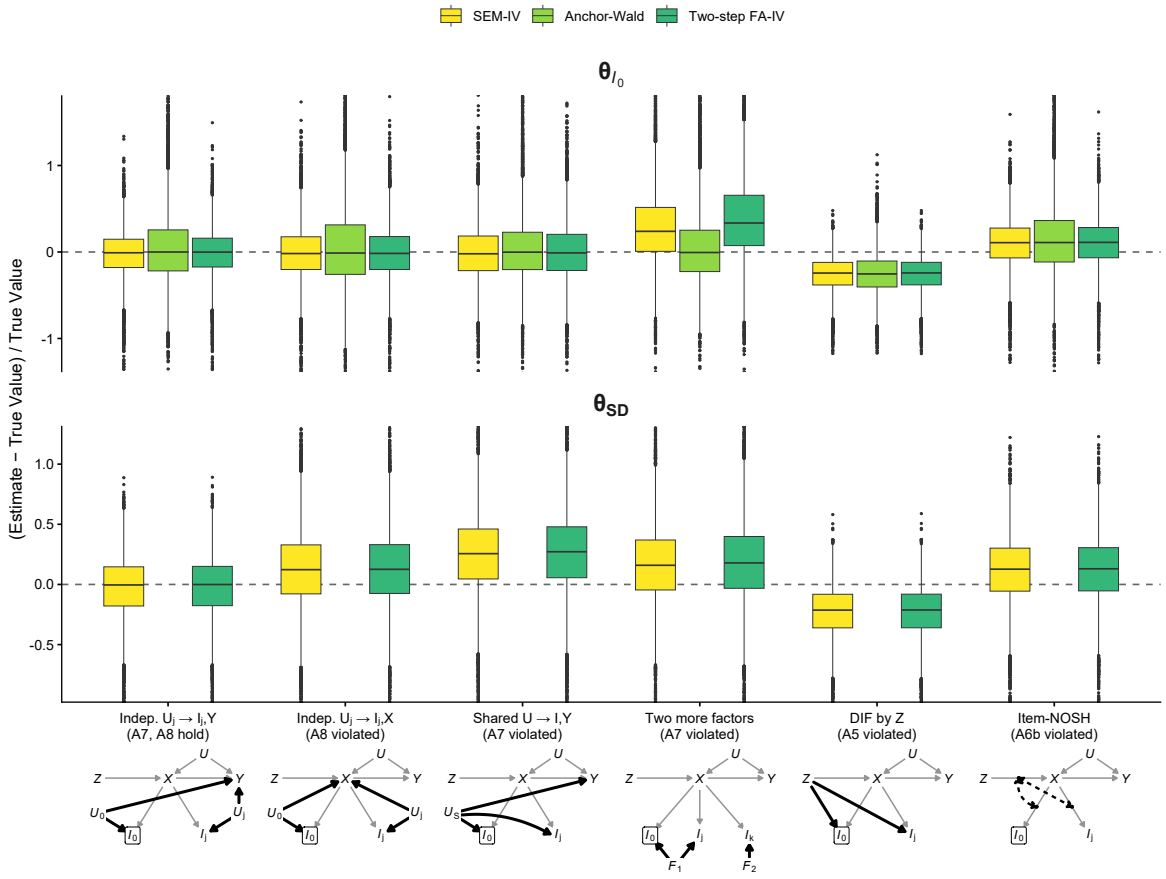
Note. Bias on the I_0 metric (top) and on the SD scale (bottom), $n = 250$. The violation sits at the ordinary items and breaks Assumption 5 (DIF by Z), 6(b) (compliance-moderated loadings), 7 (shared confounder, additional factors) or 8 (independent $U_j \to I_j, X$). Anchor-Wald has no SD-scale counterpart.

Figure 8 shows what happens when no clean anchor is available. In these scenarios, the estimators use a tenth item, also affected by the scenario's violation, as the anchor item. Because this changes the estimand, bias is reported on a relative scale. As theory predicts, anchor identification does not need Assumptions 7 and 8: Anchor-Wald stays unbiased in the first four columns. DIF by Z and Item-NOSH at the anchor break it, because anchor identification needs Assumptions 5 and 6 at the anchor.

Under independent $U_j \to I_j, X$ confounding and a shared $U \to I_j, Y$ confounder, SEM-IV and two-step FA-IV show no bias for θ_{I_0}, although both are biased in the same scenarios with a clean anchor. The reason is that every item's contamination, the anchor's included, is drawn from the same distribution. Unlike in the clean-anchor case, the item loadings are therefore correct on average relative to the anchor loading, and the multi-item estimators appear centered across replications. For any given set of items, however, they are biased, which shows in coverage that falls as n grows (Tables C6 and C7). If the anchor is contaminated differently from the rest, as in the scenario with two additional factors, where it loads on the smaller factor, the multi-item estimators are biased. Anchor-Wald does not compare the anchor with the other items and stays unbiased. For θ_{SD}, which also requires the variance of the factor, the picture is unchanged: every violation except independent $U_j \to I_j, Y$ confounding distorts estimation. Tables C6–C9 report the full set of contaminated-anchor designs at both sample sizes and on both scales.

Figure 8

Bias With a Contaminated Anchor



Note. Relative bias on the I_0 metric (top) and on the SD scale (bottom), $n = 250$. Theory predicts θ_{I_0} to be identified in every condition except DIF by Z and Item-NOSH at the anchor (Assumptions 5 and 6(b)), and θ_{SD} only in the first. The contaminated anchor's loading is drawn anew in each replication, so its target $\theta_{I_0} = \beta_Y/\lambda_0$ changes with it and bias is shown relative to each replication's true value. Anchor-Wald has no SD-scale counterpart.

Discussion

Psychological experiments do not directly manipulate a treatment variable X, which means that estimating the causal effect of X on Y is not guaranteed by design but requires additional assumptions: three classical IV assumptions that make Z a valid instrument and one assumption for the structural relationship between X and Y (NOSH). If X is latent, its scale is not intrinsic and multiple estimands may be identified. This paper developed strategies to identify two of those estimands. Via exclusion restrictions, the effect of Z on

any downstream variable, be it an indicator or the outcome, should run only via X (Assumptions 2 and 5). Via uncorrelated-heterogeneity conditions, the person-level treatment effects and item loadings must be uncorrelated with first-stage β_{Xi} (Assumptions 4(b) and 6(b)). Our simulation study violated these assumptions one at a time and demonstrated the resulting bias.

Which Estimand, and at What Cost

Because latent variables have no inherent scale, causal effects of a latent X can be expressed in two main ways: via an intervention that shifts X by one SD, or per unit on the scale of an anchor item, I_0. Both estimands require the traditional IV identification assumptions 1-4, yet they differ in the assumptions they require of the indicators. The SD strategy recovers the loadings, and hence the latent variance, from the inter-item covariance structure, so Assumptions 5–8 must hold at every indicator in the model. θ_{I_0} requires only Assumptions 5 and 6, and only at the anchor. It restores the symmetry of the design by treating one indicator exactly as a second outcome of X, reached through the instrument in the same way Y is. θ_{SD}, in contrast, demands more of the measurement model than of the outcome. With linear item response functions, the anchor estimand θ_{I_0} is therefore identified under a strict subset of the assumptions required for the scale-based estimand θ_{SD}.

The choice between the two estimands is not only a question of which assumptions one is willing to defend: whenever θ_{SD} is identified, a valid anchor necessarily exists. The decision is more about which estimand is useful. θ_{SD} is familiar but has a major problem: its unit is fixed by $\text{Var}(X)$, which depends on instrument strength.[5] A stronger Z inflates $\text{Var}(X)$ and yields a larger θ_{SD} even when the causal effect is unchanged. This matters especially in psychological experiments, where Z is only a vehicle for shifting X and is

[5] Under Assumptions 5–8, the effect can also be expressed per SD of X within a single condition, such as the control condition ($Z = 0$). This can be useful in two cases. First, if the control manipulation has a constant effect on X, for instance no effect for any participant, the SD in the control condition is that of X in the unmanipulated population, and the effect can be transported. Second, studies that share the same control manipulation $Z = 0$ can be compared even if their $Z = 1$ differs.

otherwise arbitrary (this is also why the LATE is not of interest, see Box 1). That standardized effects need care when compared across studies is well known (Pek & Flora, 2018). θ_{SD} adds a further problem: it is standardized to an artificial experimental condition, for example the spread of X that one particular sad film produces, whereas causal questions in psychology usually concern X itself. The effect is then comparable only with studies that shift X in a sufficiently similar way. The anchor estimand, too, transports to other populations only if the anchor's average loading is the same there. It does not depend on the instrument, however: given Assumptions 5 and 6 at the anchor, Z does not affect the anchor loading.

The SD estimand is the more appealing choice for a latent X that is measured with no clear anchor but with a variety of valid indicators, which differ from task to task or from study to study. Cognitive tasks with a latent X such as memory or general ability are of this kind. The anchor strategy fits the opposite case: it is appealing if few items are available and the same established items already recur across studies. Personality research is one example, where a Big Five item can serve as the anchor. The anchor strategy fits with trends that emphasize shorter scales (Allen et al., 2022; Eisele & Houben, 2026; Matthews et al., 2022), but some estimators can also use information of other items to gain efficiency, provided Assumptions 7 and 8 hold at those items.

Connection to Measurement Invariance

Identification for both estimands requires that X is measured in the same way in both conditions of Z, which is why the assumptions made here resemble familiar assumptions about measurement invariance. Measurement invariance is typically defined as a probability statement: among participants with the same value of the latent variable, the indicators must have the same joint distribution in both conditions, $(I_1, \ldots, I_J) \perp\!\!\!\perp Z \mid X$ (Meredith, 1993). This is usually discussed as a prerequisite for comparing latent variables across populations; here, only the experimental conditions $Z \in \{0, 1\}$ are compared.

Recent work has recognized that traditional measurement invariance assumptions

demand far more than most analyses require, and more than most analysts are willing to defend (Rohrer & Paulewicz, 2025; Sterner et al., 2026). They hold each indicator to be exogenous to every variable except the latent variable it measures, with loadings constant across persons. A factor model this restrictive trivially satisfies the conditions derived here, but assumes much more than IV analyses typically do. Suitable tests of measurement invariance, like multi-group SEMs, fail if any of Assumptions 5, 6 or 8 is violated, provided the other items are invariant. Yet they can also fail when the effect is identified: θ_{I_0} does not require Assumption 8 at the anchor, and heterogeneous loadings make the test of strict invariance fail even when they are unrelated to X and Z and thus satisfy Assumption 6(b). This general understanding of measurement invariance is therefore not necessary for identification, and, more consequently, a failing test does not show which assumption, could be violated. Stating measurement invariance causally gives an opportunity to explain why invariance fails and to account for it in the analysis (Sterner et al., 2024).

Rather than asking whether measurement is invariant across treatment conditions, our approach follows the causal inference tradition (Hernán & Robins, 2006; G. W. Imbens & Angrist, 1994): state the causal estimand of interest, then find the minimal requirements under which it can be estimated from data. Accurate measurement of X that is comparable across other subgroups, such as fully transportable loadings, is at best collateral. The indicators are part of the same causal system as Z, X and Y, and the assumptions they must satisfy follow from the estimand and its identification strategy. The causal inference literature states assumptions about the causal substance, independent of estimation or modeling technique, and this makes those assumptions easier to evaluate. In the running example, Assumption 6(b) can be stated with reference to features of the experimental design: it fails if participants who are moved more by the film also use the sadness rating more extremely, for instance because both depend on emotional expressivity. Stated this way, the assumption names a pre-treatment moderator that can be measured, checked and adjusted for. In contrast, assumptions that are stated about the model, such

as "constant loadings" or $(I_1, \ldots, I_J) \perp\!\!\!\perp Z \mid X$, offer no such handle.

Limitations and Future Work

The identification results presented here rest on assumptions that psychometric applications routinely make, often in a stronger form. Still, they are restrictive and should be stated carefully in practice. Two of them are linearity requirements that could be relaxed. Linearity of $X \to Y$ (Assumption 4(a)) is probably a minor concern, because the measurement error common in psychological studies tends to make nonlinear effects look linear anyway (Jacobucci & Grimm, 2020). Linearity of $X \to I_0$ (Assumption 6(a)) matters more in practice, because it restricts the anchor strategy to items that can be treated as continuous. That excludes the binary and ordinal item response models that dominate psychological measurement (Van Der Linden & Hambleton, 1997), which are nonlinear by construction. Existing alternatives trade linearity against other assumptions: they assume unconfoundedness (Schennach, 2007) or strict homogeneity (Stoetzer et al., 2025), or they require proxies that vary as richly as the latent variable they stand in for (Tchetgen et al., 2020). Extending the results presented here to IRT models, while retaining sufficient generalizability or heterogeneity, remains an open problem. One approach might be to define the item's potential outcomes as expected ratings given U_i and ϵ_{ji}, as Hartwig et al. (2023) do to accommodate binary variables.

Our simulation study had a limited scope and did not cover a range of realistic scenarios. The scenarios check the theoretical results and are no substitute for a design or power analysis. Each scenario violates a single assumption at moderate strength, with a strong instrument, a large causal effect, and high loadings. Applied studies likely face several violations at once. Weaker instruments, smaller effects, and lower loadings make every estimator less precise. The Anchor-Wald estimator is most affected, since it rests on a single item. The sample size a study needs, and the estimator that performs best in it, depend on all of these parameters jointly. A realistic power analysis therefore would need plausible values for each parameter, including the strength of every source of confounding

and the combinations in which these sources are likely to co-occur. Future work should evaluate the estimators in more realistic scenarios and address the bias–variance trade-off of adding more items: each additional item may gain efficiency but may also add confounding.

Throughout, Y was treated as unidimensional or manifest for simplicity. The framework extends directly to a latent or multidimensional Y, for example by adding an indicator of Y to the Wald ratio. However, the approach does not generalize to multidimensional treatment variables X. Because each facet can be changed separately, the treatment can be changed in several ways, which makes the estimand harder to define. If the facets of X differ in their effect on Y, it is unclear which change the causal estimand refers to, and (causal) consistency between data and estimand fails (Junker & Grosz, 2026). In our example, sadness that comes with a sense of loss may trigger a search for immediate reward, which shifts financial decisions (Lerner et al., 2013); sadness that comes with a sense of personal failure might not. A causal estimand that takes broadly defined "sadness" as X is then ill-defined. Treatment variables should therefore be sufficiently unidimensional. This makes (causal) consistency more plausible and the exclusion restrictions (Assumptions 2 and 5) easier to state.

Practical Implications for Psychological Research

The assumptions laid out above are strong and will not hold perfectly in applied settings. That is precisely why they are worth stating. A psychological experiment is constructed by the experimenter, who usually chooses both the manipulation Z and the indicators I_j from several candidates. The assumptions are therefore design targets and can be made more plausible beforehand.

IV applications in economics and epidemiology are often limited to natural experiments outside the researcher's control. In psychology, researchers can choose their Z according to desirable properties: First, the instrument Z should have a strong effect on X. Weak instruments lead at best to wide, uninformative confidence intervals, and at worst they amplify bias whenever any of the other assumptions is even slightly violated (Hernán

& Robins, 2020). Second, the exclusion restrictions on Z must hold: any causal effect of Z on Y, or on any indicator I_j, that is not fully mediated by X is problematic. Notably this is an assumption about unmeasured paths, such as indirect $Z \to U \to I_j$, outside the model. Third, Z should work through mechanisms unrelated to moderators of the $X \to Y$ relationship. A sad film depicting financial hardship moves financially strained participants most, and they are plausibly also the ones whose monetary choices react most to sadness. A sad film without financial content decouples the two and makes NOSH far more defensible.

Unfortunately, psychological methods offer no clear, empirically validated recipes for ensuring that these assumptions hold. Instead, there are many compelling examples of likely violations. Violations of IV assumptions behave differently from violations in traditional confounder-adjustment methods: they can amplify bias, so that a simple OLS estimate may be biased and still land closer to the ACE than an IV estimate in which a single assumption fails (Hernán & Robins, 2020). The exclusion restrictions, that Z affects both Y and I_j only via X, are probably the most critical assumptions; but they plague any causal inference in psychological experiments, due to their indirect nature (Eronen, 2020; Mathur, 2025). In contrast, heterogeneous effects and loadings likely matter less in practice, but allowing them comes for free.

The NOSH assumptions stated here generalize constant-effect assumptions and permit arbitrary individual-level heterogeneity in both the $Z \to X$ and the $X \to Y$ relation. This gives applied researchers a more transparent target: instead of assuming away all heterogeneity, they are asked to reason about whether the variables that make individuals more (or less) responsive to the instrument are also likely to make them more (or less) susceptible to the treatment effect. If such moderators are plausible and measured, they can be included in the treatment and outcome models, relaxing the burden placed on the unmeasured remainder of U and making the homogeneity assumption more plausible.

From a practical standpoint, this does not make the models more complicated. If the ACE is the target, the heterogeneity does not need its own parameter and can remain

in the error term. A study can still be planned under optimistic parametric conditions, for example by taking our clean baseline scenario and additionally assuming a standard LSEM, which takes effects and loadings to be constant. This makes plausible parameter values for a power analysis easier to manage. The analysis can then use the same model: in our simulation, SEM-IV, which fixes the loadings, remains unbiased when effects and loadings vary, provided the remaining assumptions hold, because the loading variance is simply absorbed into the item error.

The proposed identification strategies therefore do not require new estimation techniques: off-the-shelf software such as `lavaan` suffices, and the model settings differ from routine practice by a single scaling constraint. What may change is the target quantity, and studies that report SD scaled estimands could also report anchored estimands. The effect of sadness on financial decisions, for example, would be reported both per scale point of a single sadness rating and in units of the latent construct itself. The anchor estimand is identified under less stringent assumptions. It is not "the effect" of sadness in the abstract way that researchers may have hoped to report, but it is the effect they are better equipped to identify, defend and compare. Thus, to quote Angrist and Pischke (2009): "Sometimes You Get What You Need."

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Appendix A

Proof

The proof has two parts. Part A shows that the Wald ratio equals the ACE when X is observed. Part B recovers the ACE from the indicators, on two different scales: a standardized latent scale that pools all indicators (Strategy 1), and an anchor item metric (Strategy 2), which requires strictly fewer assumptions. Throughout, all variables and person-level coefficients are assumed to have finite second moments, so every covariance below exists.

Part A: Manifest Treatment (Assumptions 1–4)

From the parametric assumptions in the framework section, with $X_i = X_i(Z_i)$ and $Y_i = Y_i(X_i)$ by consistency, we have

$$X_i = \alpha_X + \beta_{Xi} Z_i + \zeta_{Xi} \tag{5}$$

$$Y_i = \alpha_Y + \beta_{Yi} X_i + \zeta_{Yi}. \tag{6}$$

Substituting the first into the second and taking the covariance with Z,

$$\mathrm{Cov}(Z, Y) = \mathrm{Cov}\Big(Z,\ \underbrace{\alpha_Y + \beta_{Yi}(\alpha_X + \zeta_{Xi}) + \zeta_{Yi}}_{\perp\!\!\!\perp Z}\Big) + \mathrm{Cov}(Z,\ \beta_{Yi}\beta_{Xi} Z_i) = \mathbb{E}[\beta_{Yi}\beta_{Xi}]\ \mathrm{Var}(Z).$$

β_{Xi} and ζ_{Xi} are person-level by construction, functions of (U_i, ϵ_{Xi}) alone; β_{Yi} and ζ_{Yi} are person-level functions of (U_i, ϵ_{Yi}) alone by Assumptions 2 and 4(a). Because Z is randomized (Assumption 3), it is jointly independent of all of them. Hence the first term vanishes, and the second follows from $\mathrm{Cov}(Z, AZ) = \mathbb{E}[A]\ \mathrm{Var}(Z)$ for any $A \perp\!\!\!\perp Z$, applied to $A = \beta_{Yi}\beta_{Xi}$.

The same argument applied to (5) gives $\mathrm{Cov}(Z, X) = \beta_X\ \mathrm{Var}(Z)$. Assumption 4(b)

factors $\mathbb{E}[\beta_{Yi}\beta_{Xi}] = \beta_Y\,\beta_X$, so with $\beta_X \neq 0$ by Assumption 1,

$$\beta_{\mathrm{IV}}(X) = \frac{\mathrm{Cov}(Z,Y)}{\mathrm{Cov}(Z,X)} = \frac{\beta_Y\,\beta_X\ \mathrm{Var}(Z)}{\beta_X\ \mathrm{Var}(Z)} = \beta_Y = \theta_X. \qquad \square$$

Part B: Latent Treatment

Since X is latent, $\mathrm{Cov}(Z,Y)$ is observed but $\mathrm{Cov}(Z,X)$ is not, and must be recovered from the indicators. By Assumption 6(a) each item response function $I_{ji}(x)$ is linear in x with individual-level coefficients; with $I_{ji} = I_{ji}(X_i)$ by consistency,

$$I_{ji} = \nu_j + \lambda_{ji}X_i + \zeta_{ji}, \tag{7}$$

with item intercept ν_j, average loading $\lambda_j = \mathbb{E}[\lambda_{ji}]$, loading heterogeneity $\eta_{ji} = \lambda_{ji} - \lambda_j$, and mean-zero intercept deviation ζ_{ji}. Combining the two item-specific terms into the composite residual $\delta_{ji} = \eta_{ji}X_i + \zeta_{ji}$ gives the constant-loading form

$$I_{ji} = \nu_j + \lambda_j X_i + \delta_{ji}. \tag{13}$$

Part A rested on Z being independent of the person-level coefficients β_{Xi} and β_{Yi}, by being functions of (U_i, ϵ_{Xi}) alone and Z being randomized. The same is needed on the measurement side: an item's loading and intercept must not depend on Z, neither directly nor through the realized X_i.

Lemma 1. *Under Assumptions 3, 5 and 6(a), the measurement and first-stage coefficients are jointly independent of the instrument:* $(\lambda_{ji}, \zeta_{ji}, \zeta_{Xi}, \beta_{Xi}) \perp\!\!\!\perp Z$.

λ_{ji} and ζ_{ji} are person-level functions of (U_i, ϵ_{ji}) alone by Assumptions 5 and 6(a): the first keeps Z out of the structural function of I_j, the second fixes the loading and intercept by the individual's own item response function $I_{ji}(\cdot)$ rather than by the realized X_i. [6] Because

[6] The structural intuition—Z excluded from the structural function of each item, the IV using only the Z-correlated component of each indicator—carries over to nonlinear measurement models such as item

Z is randomized (Assumption 3), it is jointly independent of all of them.

Because the scale of a latent variable is not identified, we give two strategies, targeting different estimands.

Strategy 1: Standardized Scale (All Indicators)

The target is $\theta_{\text{SD}} = \sigma_X \beta_Y$ (Equation (10)). Because it does not change when X is rescaled, no unit for X has to be fixed; fixing $\text{Var}(X) = 1$ in estimation (Appendix B) targets the same quantity. Step 1 recovers the standardized loadings from inter-item covariances, Step 2 recovers the first stage, Step 3 combines them.

Step 1: Standardized Loadings (Assumptions 7 and 8, on the representation in Equation (13)**).** For $j \neq k$,

$$\text{Cov}(I_j, I_k) = \lambda_j \lambda_k \text{ Var}(X) + \underbrace{\text{Cov}(\delta_j, \delta_k)}_{=0 \text{ by Ass. 7}} + \underbrace{\lambda_j \text{ Cov}(X, \delta_k) + \lambda_k \text{ Cov}(X, \delta_j)}_{=0 \text{ by Ass. 8}} = (\lambda_j \sigma_X)(\lambda_k \sigma_X).$$

Note that Assumption 8 constrains the composite residual, and so excludes an item whose *loading* is shifted by a cause it shares with the trait ($\text{Cov}(X_i, \eta_{ji} X_i) \neq 0$) exactly as it excludes one whose *intercept* is. With $J \geq 3$ indicators—at least three of them with $\lambda_j \neq 0$—the pairwise products identify each standardized loading $\lambda_j \sigma_X$ up to a global sign, which is fixed by orienting the scale (e.g., requiring a designated loading to be positive).

Step 2: First-Stage Covariance (Assumptions 3, 5 and 6). Substituting the first stage in Equation (5) into the item equation above and taking the covariance with Z,

$$\text{Cov}(Z, I_j) = \underbrace{\text{Cov}\Big(Z,\, \lambda_{ji}(\alpha_X + \zeta_{Xi})\Big)}_{=0} + \text{Cov}(Z,\, \lambda_{ji} \beta_{Xi} Z_i) + \underbrace{\text{Cov}(Z,\, \zeta_{ji})}_{=0} = \mathbb{E}[\lambda_{ji} \beta_{Xi}] \text{ Var}(Z),$$

the outer terms vanishing by Lemma 1 and the middle one by the same $A \perp\!\!\!\perp Z$ argument as in Part A. Assumption 6(b), $\text{Cov}(\lambda_{ji}, \beta_{Xi}) = 0$, factors $\mathbb{E}[\lambda_{ji} \beta_{Xi}] = \lambda_j \, \beta_X$, so with

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response theory, but the covariance argument does not: both steps below rest on Assumption 6(a). Identification in such models requires its own, typically likelihood-based, proof.

$\mathrm{Cov}(Z, X) = \beta_X \mathrm{Var}(Z)$ from Part A,

$$\mathrm{Cov}(Z, I_j) = \lambda_j \mathrm{Cov}(Z, X).$$

The argument needs Assumption 6(b) only at the indicator used in Step 3; an SEM that fits all J indicators needs it at each of them.

Step 3: Combine. The Wald ratio $\beta_{\mathrm{IV}}(I_j)$ is observed, and Step 1 identifies the standardized loading $\lambda_j \sigma_X$. With $\mathrm{Cov}(Z, Y) = \mathbb{E}[\beta_{Yi}\beta_{Xi}] \mathrm{Var}(Z)$ from Part A, Assumption 4(b) factoring the numerator, and Step 2 for the denominator, for any indicator with $\lambda_j \neq 0$,

$$\lambda_j \sigma_X \, \beta_{\mathrm{IV}}(I_j) = \lambda_j \sigma_X \frac{\mathbb{E}[\beta_{Yi}\beta_{Xi}]}{\mathbb{E}[\lambda_{ji}\beta_{Xi}]} = \lambda_j \sigma_X \frac{\beta_Y \, \beta_X}{\lambda_j \, \beta_X} = \sigma_X \beta_Y = \theta_{\mathrm{SD}}. \qquad \square$$

In practice the simultaneous SEM estimator pools all J indicators rather than using any single one.

Strategy 2: Anchor Item

The anchor strategy designates one indicator I_0 and reports the effect on its metric, so it never recovers a loading: the anchor is treated as just another outcome of X, reached through the instrument exactly as Y is. Its item response function is Equation (7) with $j = 0$,

$$I_{0i} = \nu_0 + \lambda_{0i} X_i + \zeta_{0i},$$

and the anchor metric of Equation (11) is defined through $\lambda_0 = \mathbb{E}[\lambda_{0i}]$. This is nonzero, because an item unrelated to X could not serve as an anchor. Fixing the anchor's loading to 1 in estimation places X on that metric. Unlike λ_j in Strategy 1, λ_0 itself is never estimated.

Applying Step 2 to the single item I_0 and combining with $\mathrm{Cov}(Z, Y)$ from Part A,

$$\beta_{\mathrm{IV}}(I_0) = \frac{\mathrm{Cov}(Z, Y)}{\mathrm{Cov}(Z, I_0)} = \frac{\mathbb{E}[\beta_{Yi}\beta_{Xi}]}{\mathbb{E}[\lambda_{0i}\beta_{Xi}]} \overset{\text{Ass. 4(b), 6(b)}}{=} \frac{\beta_Y}{\lambda_0} = \theta_{I_0},$$

the ACE per unit of the average anchor metric. □

Appendix B

Estimation

Estimation With a Manifest Treatment

The Wald ratio $\beta_{\text{IV}}(X)$ in Equation (9) is typically estimated by two-stage least squares, but it can equivalently be estimated within a SEM framework.

Two-Stage Least Squares

In the first stage, the treatment X is regressed on the instrument Z to isolate the variation in X that is induced by the instrument, yielding predicted values $\hat{X}$. In the second stage, the outcome Y is regressed on these predicted values. The coefficient of $\hat{X}$ in the second stage is the estimator of the causal effect.

Two-stage least squares can be fit with `ivreg(Y ~ X | Z)` from the `ivreg` package (Fox et al., 2026). The simulation results report its default confidence interval, which uses t quantiles with $n - 2$ degrees of freedom. The standard error is the conventional two-stage least squares standard error, which assumes homoskedastic residuals (Angrist & Pischke, 2009). The residual variance $\hat{\sigma}^2 = \sum_i \hat{e}_i^2/(n-2)$ uses the structural residuals $\hat{e}_i = Y_i - \hat{\alpha} - \hat{b} \cdot X_i$, with $\hat{b} = \widehat{\beta}_{\text{IV}}(X)$ and $\hat{\alpha}$ the estimated intercept. These residuals use the observed treatment X_i, not its first-stage prediction $\hat{X}_i$; this is the adjustment that running the two regressions by hand misses. With a single instrument, the standard error reduces to

$$\widehat{\text{SE}}\big(\widehat{\beta}_{\text{IV}}(X)\big) = \frac{\hat{\sigma}}{\sqrt{n}\,\hat{\sigma}_X\,|\hat{\rho}_{Z,X}|},$$

where n is the sample size, $\hat{\sigma}_X$ the sample standard deviation of the treatment, and $\hat{\rho}_{Z,X}$ the sample correlation between instrument and treatment, that is, the strength of the first stage. A stronger first stage therefore gives a smaller standard error.

Structural Equation Modeling

In psychology, it is often more convenient to specify the IV model as a path model using the `lavaan` package. This approach naturally extends to the latent variable settings

discussed below. The instrument Z predicts X (relevance), while the exclusion restriction is enforced by omitting the direct path from Z to Y. The unobserved confounding between X and Y is modeled explicitly as a covariance between the residual errors of X and Y, $\text{Cov}(\zeta_{Xi}, \zeta_{Yi})$ in Equations (5) and (6) (the `Y ~~ X` operator in `lavaan`); this residual covariance captures the endogeneity caused by unmeasured confounders such as financial strain.

```
# IV estimation with manifest variables
iv_model <- '
# 1. First Stage: Instrument Z predicts Treatment X
X ~ Z
# 2. Structural Equation: Treatment X affects Outcome Y
Y ~ X
# 3. Account for Endogeneity (Confounding)
# Allow residuals of X and Y to correlate to capture U
Y ~~ X
'
```

Estimation With a Latent Treatment

The three estimators compared in the simulation all operate on the linear measurement model of Equation (13), $I_{ji} = \nu_j + \lambda_j X_i + \delta_{ji}$. The two-step factor-score IV and the simultaneous SEM-IV can target either θ_{SD} or θ_{I_0}, depending on how the scale of the latent treatment is set: with the loading of I_0 fixed to one, the structural coefficient is on the anchor metric and estimates θ_{I_0}; with the factor variance fixed to one, it estimates θ_{SD}. The anchor-Wald estimator targets θ_{I_0} exclusively.

Two-Step Estimation (Factor Scores)

In the first approach, the latent variable is operationalized by estimating a Confirmatory Factor Analysis (CFA) model and extracting factor scores. These scores are

then treated as a manifest variable in a standard two-stage least squares regression. While computationally simple, this method treats the factor scores as observed in the second stage. The standard errors are the two-stage least squares standard errors of the manifest case with the scores in place of X. The measurement error of the scores enters the second-stage residual and is therefore reflected in these standard errors.

```
# Step 1: Fit the measurement model (CFA)
cfa_model <- paste0("X =~ ", paste(itemNames, collapse = " + "))
fit_cfa <- cfa(cfa_model, data = data, std.lv = TRUE, estimator = "MLR")
# Step 2: Extract Bartlett factor scores
scores <- lavPredict(fit_cfa, method = "Bartlett")
# Step 3: Estimate IV regression using scores as observed treatment
fit_iv <- ivreg(y ~ scores | z, data = data)
```

The code shows the θ_{SD} version (`std.lv = TRUE`); for θ_{I_0}, set `std.lv = FALSE` to fix the loading of I_0. The θ_{SD} estimate and its confidence interval equal the θ_{I_0} estimate and interval multiplied by the CFA estimate of the SD of X (the square root of the estimated factor variance), so the interval treats that SD as known.

Regression scores, the default in `lavaan` and most software, shrink toward the mean, so Z moves them less than it moves the factor and the IV estimate is inflated; Bartlett scores must therefore be requested explicitly (`method = "Bartlett"` in `lavPredict()`).

Simultaneous Estimation (SEM-IV)

Another approach is to estimate the measurement model and the structural IV parameters simultaneously within a single SEM. This fully accounts for measurement error and parameter uncertainty.

In `lavaan`, this is achieved by defining the latent variable X via its indicators and specifying the IV structure (instrument relevance and endogeneity) in the same model string.

```
# IV estimation with latent treatment
iv_sem_model <- paste0(
# 1. Measurement Model
"X =~ ", paste(itemNames, collapse = " + "), "
# 2. First Stage: Instrument Z predicts Latent X
X ~ Z
# 3. Second Stage: Latent X predicts Outcome Y
Y ~ X
# 4. Model Endogeneity
# Allow residuals of Latent X and Y to covary
X ~~ Y"
)
# 5. Fit by robust maximum likelihood
fit <- sem(iv_sem_model, data = data, estimator = "MLR")
```

We fit both the SEM and the CFA of the two-step estimator by robust maximum likelihood (`estimator = "MLR"`), which gives the maximum likelihood point estimates with robust (Huber–White) standard errors. The structural coefficient inherits the scale of the latent variable X. With the loading of I_0 fixed to one, it estimates θ_{I_0}. The SD-scaled effect θ_{SD} is retrieved from the `std.lv` column of `lavaan`'s standardized output (`standardizedSolution(fit, type = "std.lv")`), which multiplies the coefficient by the model-implied SD of X. Its delta-method confidence interval accounts for the fact that this SD is estimated. The `std.all` column would additionally scale it by the SD of Y.

Anchor-Wald Estimator

The anchor-Wald estimator uses I_0 directly as the endogenous treatment variable in a just-identified IV regression:

$$\widehat{\beta}_{\mathrm{IV}}(I_0) = \frac{\widehat{\mathrm{Cov}}(Z, Y)}{\widehat{\mathrm{Cov}}(Z, I_0)}.$$

Its population counterpart $\beta_{\mathrm{IV}}(I_0)$ equals $\beta_Y/\lambda_0 = \theta_{I_0}$ under Assumptions 1–4, with 5 and 6 at the anchor (Appendix A, Strategy 2). When $\lambda_0 = 1$, the estimator is on the same scale as the latent treatment; more generally, it estimates the IV effect per one-unit increase in the anchor metric. This is the same metric used by anchored CFA or SEM specifications in which the loading of I_0 is fixed to one, that is, it targets θ_{I_0} directly.

The anchor-Wald estimator does not use the remaining indicators at all, which makes it a transparent benchmark: it avoids misspecification of the full measurement model, but it relies entirely on the validity of the chosen anchor. Classical measurement error in I_0 that is independent of Z only weakens the first stage and reduces precision, leaving the population ratio unchanged. The simulation fits the anchor-Wald estimator with `ivreg(Y ~ I0 | Z)` and reports the two-stage least squares confidence interval of the manifest case, with the observed anchor I_0 in place of X. Its standard error therefore shrinks with the strength of the anchor's first stage, $|\hat{\rho}_{Z,I_0}|$.

Appendix C

Simulation

Data-Generating Process

Data come from a structural model with a reflective measurement block. All functions are linear, but the slopes vary across persons, so the simulation never assumes a constant effect. The baseline model is

$$(\beta_{Xi}, \beta_{Yi}) \sim \mathcal{N}\left(\begin{pmatrix}\beta_X \\ \beta_Y\end{pmatrix}, \begin{pmatrix}\sigma^2_{\beta_X} & \rho\,\sigma_{\beta_X}\sigma_{\beta_Y} \\ \rho\,\sigma_{\beta_X}\sigma_{\beta_Y} & \sigma^2_{\beta_Y}\end{pmatrix}\right),$$

$$X_i = \beta_{Xi} Z_i + \pi_X U_i + \epsilon_{Xi},$$

$$Y_i = \beta_{Yi} X_i + \pi_Y U_i + \epsilon_{Yi},$$

with U_i, ϵ_{Xi} and ϵ_{Yi} standard normal and the population values in Table C1. Z is assigned by complete randomization, with exactly $n/2$ persons per arm. ρ is the correlation between the slopes β_{Xi} and β_{Yi}; $\rho \neq 0$ violates Assumption 4(b).

Ten ordinary reflective items measure X,

$$I_{ji} = \nu_j + \lambda_{ji} X_i + \zeta_{ji}, \qquad \lambda_{ji} = \lambda_j + \eta_{ji}, \qquad \eta_{ji} = \lambda_j \left(e_{ji} + \kappa_{Cj}\, c_{Xi}\right), \qquad j = 1, \ldots, 10,$$

with $e_{ji} \sim \mathcal{N}(0, \mathrm{CV}^2)$, so that the loading heterogeneity is proportional to the average loading, $\mathrm{SD}(\lambda_{ji}) = \mathrm{CV}\,\lambda_j$. The second term lets the loading depend on the standardized compliance $c_{Xi} = (\beta_{Xi} - \beta_X)/\sigma_{\beta_X}$. It is zero in the baseline ($\kappa_{Cj} = 0$); setting $\kappa_{Cj} \neq 0$ violates Assumption 6(b), because each loading then covaries with compliance, $\mathrm{Cov}(\lambda_{ji}, \beta_{Xi}) = \kappa_{Cj}\lambda_j\sigma_{\beta_X}$. Intercepts, residual SDs and standardized average loadings are drawn once per item and data set. Every data set also contains a clean anchor I_{0i} of the same form without the compliance term, with a fixed standardized loading and a fixed intercept. Table C1 shows the parameters of the simulation.

Table C1

Population parameters of the baseline data-generating process

Symbol	Value	Role
β_Y	1.5	ACE: mean of β_{Yi} (Y on X)
β_X	1.5	Mean of β_{Xi} (X on Z)
π_X	2	Confounder $U \to X$
π_Y	2.5	Confounder $U \to Y$
σ_{β_Y}	1.0	SD of β_{Yi}: heterogeneity in $X \to Y$
σ_{β_X}	0.5	SD of β_{Xi}: heterogeneity in $Z \to X$
ρ	0 (.4 in NOSH violated)	Correlation of β_{Xi} and β_{Yi} (Assumption 4(b))
$\sigma_{\epsilon_X}, \sigma_{\epsilon_Y}$	1, 1	Residual SDs of X and Y
λ_0	0.475	Average anchor loading (standardized loading .75, fixed; 0.364 under independent $U_j \to I_j, X$)
ν_0	0	Anchor intercept (fixed)
SD(ζ_0)	1	Anchor residual SD (reliability $\approx$ .56)
CV	0.12	Loading heterogeneity, SD(λ_{ji})/λ_j, all items
Mean, SD	0.7, 0.1	Standardized loadings, ordinary items (truncated to $[.20, .85]$)
SD(ζ_j)	$\mathcal{U}(.5, 1.5)$	Residual SDs, ordinary items
K	10	Number of ordinary items
π_j, κ_j	$\mathcal{N}(.7, .21)$, $\mathcal{N}(.21, .14)$	Effect of the contaminating variable on the intercept and loading of item j
π_{Xj}	$2/\sqrt{10}$	Independent $U_j \to X$ design only
π_{Yj}	$2/\sqrt{10}$	Independent $U_j \to Y$ design only
π_{YS}	2	Shared confounder design only ($U_S \to Y$)
$\Delta\nu_j, \Delta\lambda_j$	.175, .03	DIF by Z only: intercept shift and shift in the standardized loading
κ_{Cj}	$\mathcal{N}(-.3, .1)$	Item-NOSH only: loading moderated by compliance, relative to λ_j (Assumption 6(b))

Note. ACE = average causal effect; NOSH = no simultaneous heterogeneity; CV = coefficient of variation; DIF = differential item functioning.

Scenarios

Besides the baseline scenario, seven scenarios are generated, each adds a distinct violation and leaves everything else at the baseline. The terms that are on in each scenario, and the equation they enter, are listed next to the scenario definitions in `functions/run_simulation.R`; the general model they plug into is in `functions/simulate_data.R`. Table 2 describes each scenario in words.

The clean-anchor designs use the clean anchor and nine ordinary items. Here, only the ordinary items are affected by a violation. In the contaminated-anchor designs, the clean anchor is dropped and the tenth ordinary item, which carries the same contamination, becomes I_0, with the same nine ordinary items as before.

The four designs with a contaminating variable use the same effect on the items, so the affected items have comparable loadings. The contaminating variable shifts the item intercept by π_j and moderates the loading by κ_j. Only where it also acts on X, under independent $U_j \to I_j, X$, does the item residual correlate with X, and then through both channels.

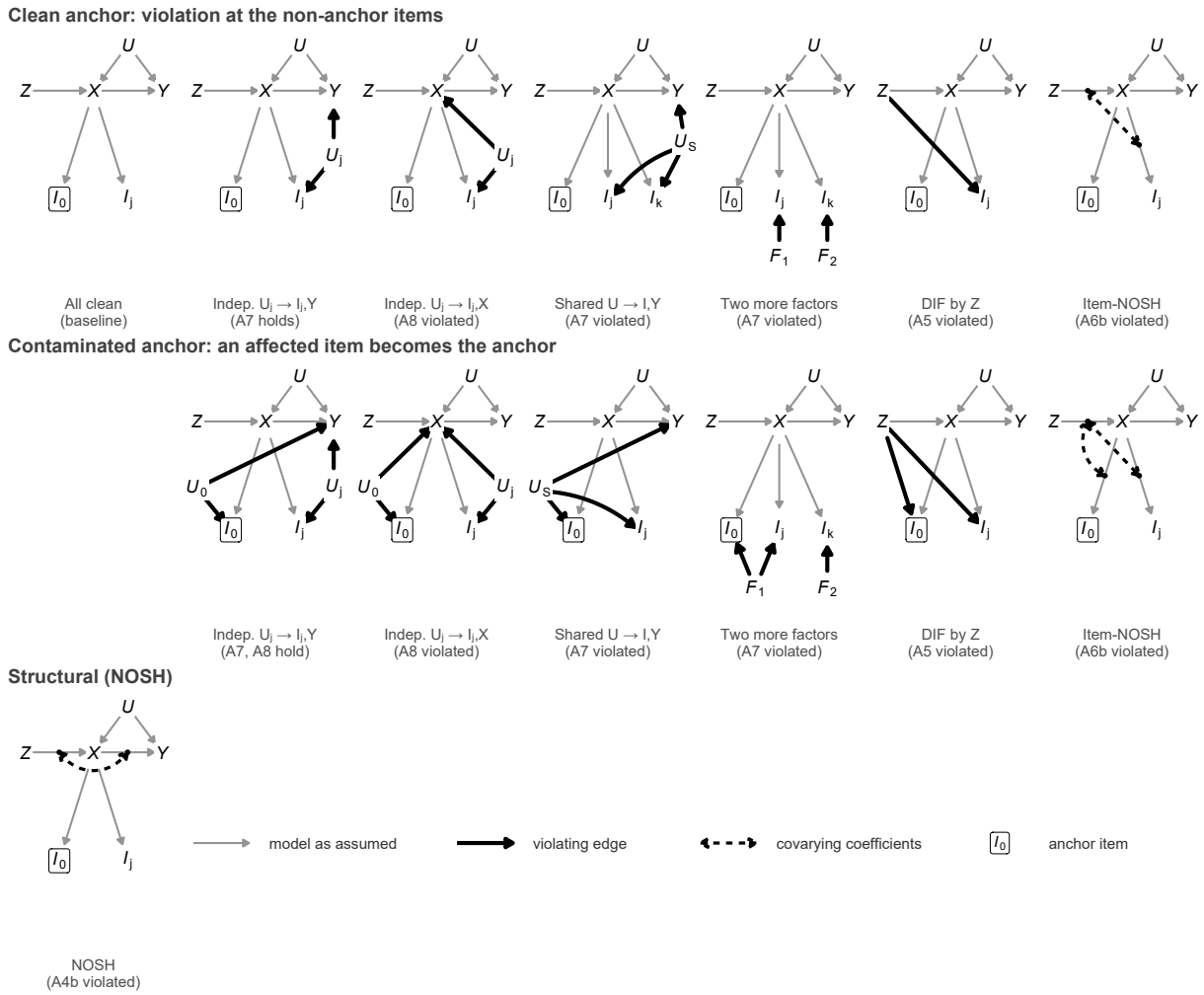
Performance Measures and Intervals

All tables report median bias, RMSE and empirical 95% coverage over 10,000 replications. Coverage is the share of replications in which the confidence interval contains the true value. A replication fails for SEM-IV when the model does not converge, and for two-step FA-IV when the CFA does not converge or returns a negative variance estimate. SEM-IV is not screened for negative variances. Of the 140,000 replications per estimator and sample size, SEM-IV failed in 2 at $n = 250$ and 0 at $n = 2{,}000$, two-step FA-IV in 0 and 0. Failed replications are left out of that estimator's summaries.

Confidence intervals are the software defaults from `ivreg` for Anchor-Wald and two-step FA-IV; and robust ML from `lavaan` for SEM-IV. On the SD scale, SEM-IV uses the delta-method interval of the `std.lv` coefficient, which accounts for the estimated SD of X. The two-step interval comes from the `std.lv = TRUE` fit and treats the SD of X as

Figure C1

Schematics of the Simulation Scenarios



Note. Each panel is one scenario. U_j denotes confounding that is idiosyncratic to an item, whereas U_S denotes a confounder that the items share between them. A NOSH violation is a covariance between coefficients rather than between variables, so it is drawn as a dashed arc joining two edges. F_1 and F_2 are the two additional factors; U_0 is the confounder of the contaminated anchor.

known. At $n = 250$ the `ivreg` intervals are slightly conservative, most visibly for Anchor-Wald, whose first stage is the weakest.

The tables report the median because the just-identified Wald ratio has no finite moments: a small share of replications with a weak first stage inflates the mean and the RMSE without moving the bulk of the distribution. The RMSE columns stay so that this dispersion remains visible. At $n = 250$ this shows most for Anchor-Wald: in a few replications the anchor barely differs between the arms, and these replications make up nearly all of its RMSE. Anchor-Wald has no SD-scale counterpart, so only SEM-IV and two-step FA-IV appear in the SD tables.

Performance Results

The designs with a clean anchor come first, on the I_0 metric (Tables C2 and C3) and on the SD scale (Tables C4 and C5), each at $n = 250$ and $n = 2{,}000$. In each table the rule after the third row separates the scenarios in which theory predicts the same verdict for both estimands from those in which only θ_{I_0} survives.

In the contaminated-anchor designs, the first affected item serves as I_0. Figure C2 summarizes the median bias at $n = 2{,}000$; the full results are on the I_0 metric (Tables C6 and C7) and on the SD scale (Tables C8 and C9). On the I_0 metric, the target is β_Y/λ_0 of the contaminated anchor, whose loading is drawn anew in each replication, so bias and RMSE are reported relative to each replication's true value.

Table C2

Performance on the I_0 metric with a clean anchor, $n = 250$

	SEM-IV			Anchor-Wald			Two-step FA-IV		
Scenario	Bias	RMSE	Cov.	Bias	RMSE	Cov.	Bias	RMSE	Cov.
All clean	0.010	0.779	.964	0.016	1.009	.975	0.016	0.782	.952
Indep. $U_j \to I_j, Y$	0.073	0.886	.958	-0.019	1.634	.976	-0.003	0.869	.954
NOSH violated	0.267	0.783	.927	0.281	1.076	.978	0.272	0.787	.903
Indep. $U_j \to I_j, X$	0.401	1.735	.963	0.033	10.892	.976	0.437	2.268	.952
DIF by Z	-0.601	0.934	.893	0.007	1.040	.975	-0.603	0.934	.886
Shared $U \to I, Y$	1.419	2.145	.710	0.006	2.388	.975	1.730	2.564	.617
Two more factors	0.923	1.807	.874	-0.011	1.062	.975	1.258	2.308	.792
Item-NOSH violated	0.438	0.989	.916	-0.006	1.040	.976	0.534	1.062	.877

Note. Bias = median bias; RMSE = root-mean-square error; Cov. = empirical coverage of the 95% confidence interval; Indep. = independent; DIF = differential item functioning; NOSH = no simultaneous heterogeneity. True value $\theta_{I_0} = 3.155$ (4.117 under independent $U_j \to I_j, X$).

Table C3

Performance on the I_0 metric with a clean anchor, $n = 2{,}000$

	SEM-IV			Anchor-Wald			Two-step FA-IV		
Scenario	Bias	RMSE	Cov.	Bias	RMSE	Cov.	Bias	RMSE	Cov.
All clean	-0.006	0.252	.951	-0.002	0.309	.954	-0.006	0.252	.941
Indep. $U_j \to I_j, Y$	0.082	0.285	.936	-0.003	0.333	.954	-0.002	0.269	.943
NOSH violated	0.274	0.364	.779	0.277	0.418	.872	0.275	0.365	.753
Indep. $U_j \to I_j, X$	0.382	0.571	.846	-0.005	0.502	.954	0.406	0.589	.819
DIF by Z	-0.606	0.653	.202	-0.003	0.306	.954	-0.609	0.655	.178
Shared $U \to I, Y$	1.397	1.705	.092	0.006	0.331	.955	1.701	2.079	.051
Two more factors	0.918	1.269	.308	0.010	0.305	.958	1.256	1.660	.164
Item-NOSH violated	0.448	0.540	.636	-0.001	0.307	.955	0.541	0.629	.516

Note. Bias = median bias; RMSE = root-mean-square error; Cov. = empirical coverage of the 95% confidence interval; Indep. = independent; DIF = differential item functioning; NOSH = no simultaneous heterogeneity. True value $\theta_{I_0} = 3.155$ (4.117 under independent $U_j \to I_j, X$).

Table C4

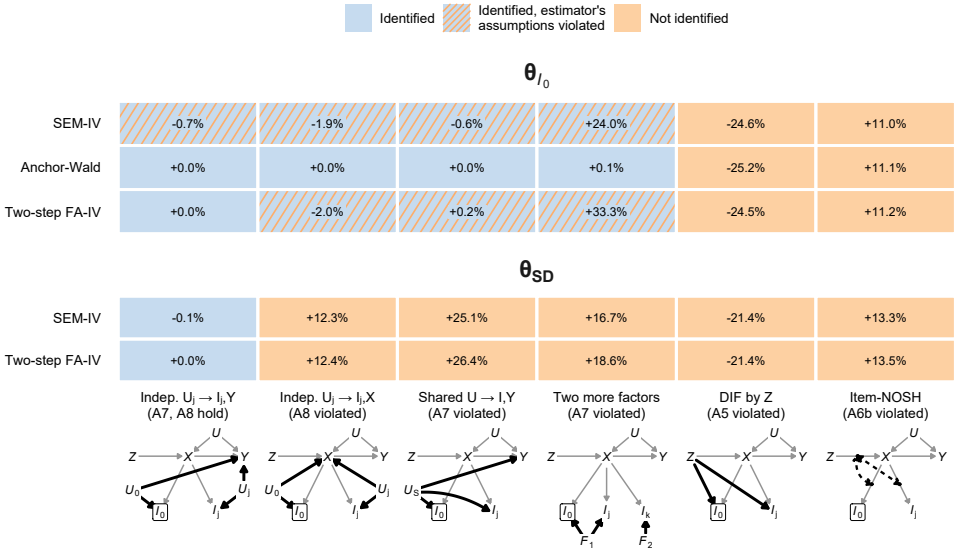
Performance on the SD scale with a clean anchor, $n = 250$

	SEM-IV			Two-step FA-IV		
Scenario	Bias	RMSE	Cov.	Bias	RMSE	Cov.
All clean	-0.005	0.900	.963	0.000	0.901	.948
Indep. $U_j \to I_j, Y$	-0.023	0.988	.964	-0.012	0.991	.949
NOSH violated	0.288	0.894	.930	0.292	0.897	.903
Indep. $U_j \to I_j, X$	0.485	1.973	.955	0.496	2.472	.949
DIF by Z	-0.706	1.086	.887	-0.702	1.084	.868
Shared $U \to I, Y$	0.775	1.471	.858	0.837	1.553	.833
Two more factors	0.528	1.372	.933	0.604	1.522	.927
Item-NOSH violated	0.372	1.052	.932	0.389	1.061	.913

Note. Bias = median bias; RMSE = root-mean-square error; Cov. = empirical coverage of the 95% confidence interval; Indep. = independent; DIF = differential item functioning; NOSH = no simultaneous heterogeneity. True value $\theta_{\text{SD}} = 3.58$ (4.67 under independent $U_j \to I_j, X$).

Figure C2

Median Bias With a Contaminated Anchor



Note. $n = 2{,}000$. Cells show the relative median bias, the median of (estimate − true value) / true value. Shading marks whether theory predicts the estimand to be identified; hatching, as in Figure 5, marks an identified estimand whose estimator relies on assumptions the scenario violates. The graph under each column marks the violating path (black); the boxed node is the anchor.

Table C5

Performance on the SD scale with a clean anchor, $n = 2{,}000$

	SEM-IV			Two-step FA-IV		
Scenario	Bias	RMSE	Cov.	Bias	RMSE	Cov.
All clean	-0.006	0.294	.950	-0.005	0.294	.935
Indep. $U_j \to I_j, Y$	-0.006	0.311	.956	-0.002	0.312	.939
NOSH violated	0.312	0.419	.788	0.313	0.419	.744
Indep. $U_j \to I_j, X$	0.484	0.687	.820	0.487	0.689	.804
DIF by Z	-0.707	0.762	.211	-0.706	0.761	.172
Shared $U \to I, Y$	0.772	0.942	.424	0.826	1.006	.367
Two more factors	0.550	0.753	.654	0.623	0.837	.591
Item-NOSH violated	0.397	0.514	.749	0.408	0.524	.705

Note. Bias = median bias; RMSE = root-mean-square error; Cov. = empirical coverage of the 95% confidence interval; Indep. = independent; DIF = differential item functioning; NOSH = no simultaneous heterogeneity. True value $\theta_{\text{SD}} = 3.58$ (4.67 under independent $U_j \to I_j, X$).

Table C6

Performance on the I_0 metric with a contaminated anchor (an affected item serves as I_0), $n = 250$

	SEM-IV			Anchor-Wald			Two-step FA-IV		
Scenario	Bias	RMSE	Cov.	Bias	RMSE	Cov.	Bias	RMSE	Cov.
Indep. $U_j \to I_j, Y$	-0.009	0.283	.967	0.000	7.322	.962	-0.001	0.290	.938
Indep. $U_j \to I_j, X$	-0.018	0.368	.970	-0.012	11.091	.952	-0.017	0.375	.958
Shared $U \to I, Y$	-0.021	0.338	.903	-0.002	23.271	.967	-0.010	0.361	.881
Two more factors	0.239	0.592	.930	-0.006	3.322	.959	0.335	0.751	.823
DIF by Z at the anchor	-0.243	0.326	.789	-0.254	0.347	.763	-0.242	0.326	.784
Item-NOSH at the anchor	0.108	0.302	.941	0.109	2.319	.979	0.111	0.305	.914

Note. Bias = median bias; RMSE = root-mean-square error; Cov. = empirical coverage of the 95% confidence interval; Indep. = independent; DIF = differential item functioning; NOSH = no simultaneous heterogeneity. Bias and RMSE are relative to each replication's true value $\theta_{I_0} = \beta_Y / \lambda_0$ of the contaminated anchor, which changes with the loading drawn for that item.

Table C7

Performance on the I_0 metric with a contaminated anchor (an affected item serves as I_0), $n = 2{,}000$

	SEM-IV			Anchor-Wald			Two-step FA-IV		
Scenario	Bias	RMSE	Cov.	Bias	RMSE	Cov.	Bias	RMSE	Cov.
Indep. $U_j \to I_j, Y$	-0.007	0.090	.947	0.000	0.137	.956	0.000	0.090	.928
Indep. $U_j \to I_j, X$	-0.019	0.113	.895	0.000	0.185	.954	-0.020	0.115	.872
Shared $U \to I, Y$	-0.006	0.190	.628	0.000	0.122	.957	0.002	0.210	.579
Two more factors	0.240	0.366	.470	0.001	0.135	.956	0.333	0.477	.275
DIF by Z at the anchor	-0.246	0.257	.040	-0.252	0.273	.152	-0.245	0.256	.034
Item-NOSH at the anchor	0.110	0.144	.754	0.111	0.182	.875	0.112	0.147	.715

Note. Bias = median bias; RMSE = root-mean-square error; Cov. = empirical coverage of the 95% confidence interval; Indep. = independent; DIF = differential item functioning; NOSH = no simultaneous heterogeneity. Bias and RMSE are relative to each replication's true value $\theta_{I_0} = \beta_Y / \lambda_0$ of the contaminated anchor, which changes with the loading drawn for that item.

Table C8

Performance on the SD scale with a contaminated anchor (an affected item serves as I_0), $n = 250$

	SEM-IV			Two-step FA-IV		
Scenario	Bias	RMSE	Cov.	Bias	RMSE	Cov.
Indep. $U_j \to I_j, Y$	-0.015	0.985	.964	-0.002	0.989	.947
Indep. $U_j \to I_j, X$	0.574	1.951	.953	0.587	1.968	.948
Shared $U \to I, Y$	0.916	1.603	.839	0.972	1.678	.821
Two more factors	0.570	1.450	.930	0.640	1.576	.925
DIF by Z at the anchor	-0.763	1.120	.865	-0.759	1.118	.840
Item-NOSH at the anchor	0.454	1.106	.923	0.466	1.113	.902

Note. Bias = median bias; RMSE = root-mean-square error; Cov. = empirical coverage of the 95% confidence interval; Indep. = independent; DIF = differential item functioning; NOSH = no simultaneous heterogeneity. True value $\theta_{\mathrm{SD}} = 3.58$ (4.67 under independent $U_j \to I_j, X$).

Table C9

Performance on the SD scale with a contaminated anchor (an affected item serves as I_0), $n = 2{,}000$

	SEM-IV			Two-step FA-IV		
Scenario	Bias	RMSE	Cov.	Bias	RMSE	Cov.
Indep. $U_j \to I_j, Y$	-0.005	0.309	.954	-0.001	0.309	.939
Indep. $U_j \to I_j, X$	0.575	0.763	.775	0.578	0.765	.763
Shared $U \to I, Y$	0.898	1.067	.336	0.944	1.122	.291
Two more factors	0.597	0.805	.623	0.664	0.883	.565
DIF by Z at the anchor	-0.766	0.815	.148	-0.764	0.814	.112
Item-NOSH at the anchor	0.474	0.583	.683	0.482	0.589	.645

Note. Bias = median bias; RMSE = root-mean-square error; Cov. = empirical coverage of the 95% confidence interval; Indep. = independent; DIF = differential item functioning; NOSH = no simultaneous heterogeneity. True value $\theta_{\text{SD}} = 3.58$ (4.67 under independent $U_j \to I_j, X$).